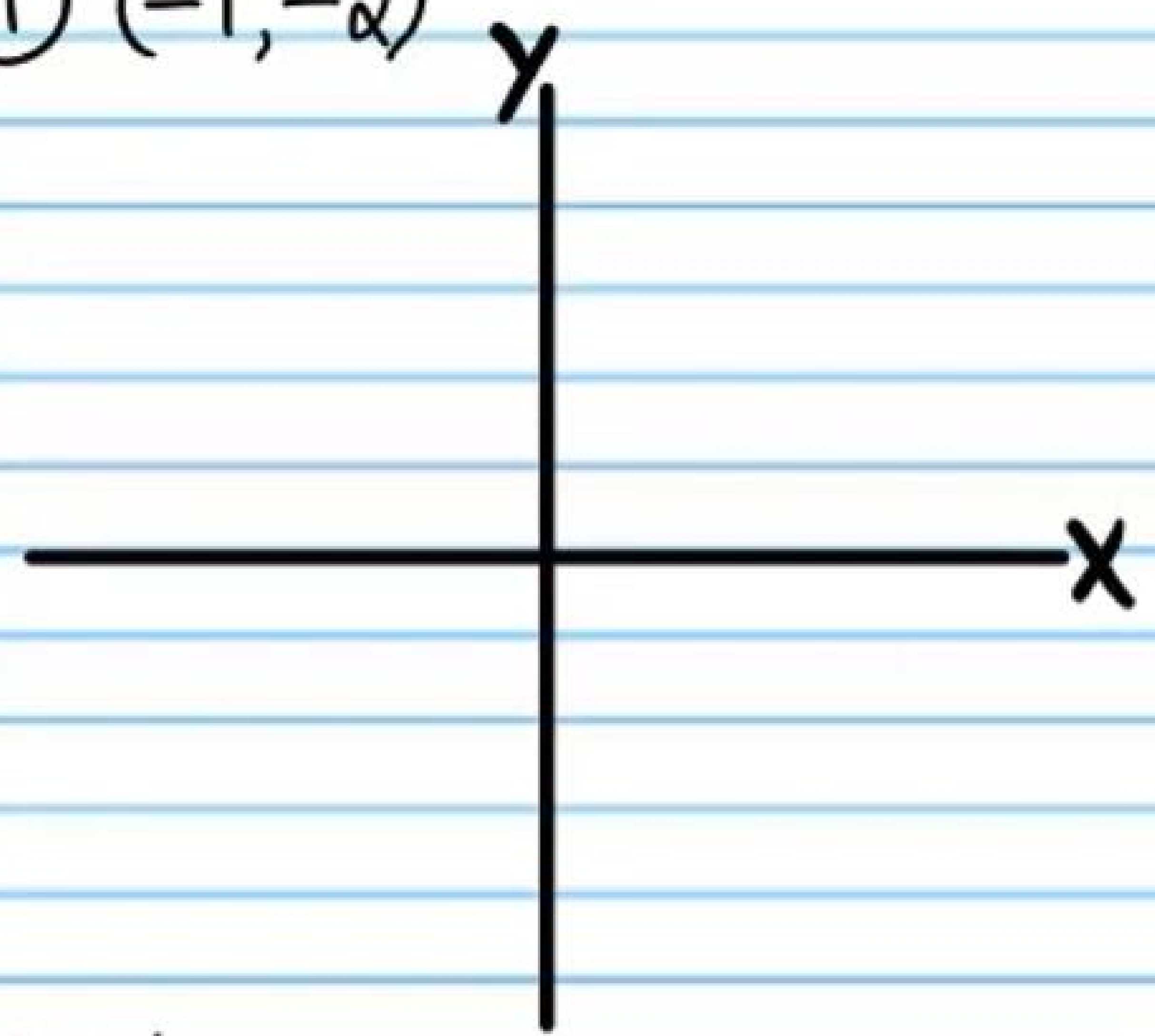
Name
Date
Course

Test I

I) Write your name, the date, and the course title (i.e. Trigonometry).
II) Show all work.
III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for this test (be sure that it is in degree mode).
V) Each problem is worth four points for a total of 100 points.

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

① $(-1,-2)$



i) $\sin\theta$

ii) $\cos\theta$

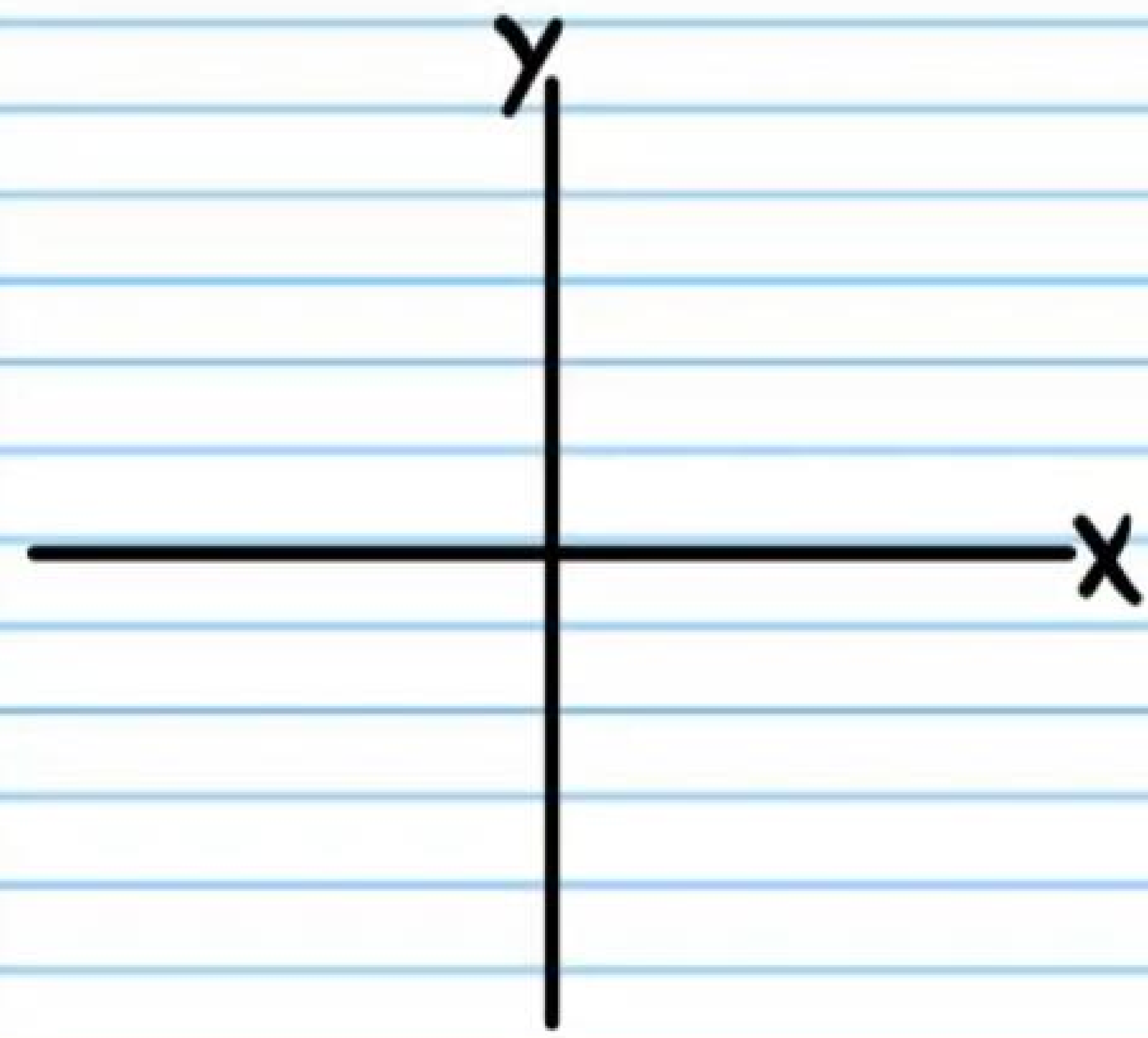
iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

② $(1,-\sqrt{3})$



i) $\sin\theta$

ii) $\cos\theta$

iii) $\tan\theta$

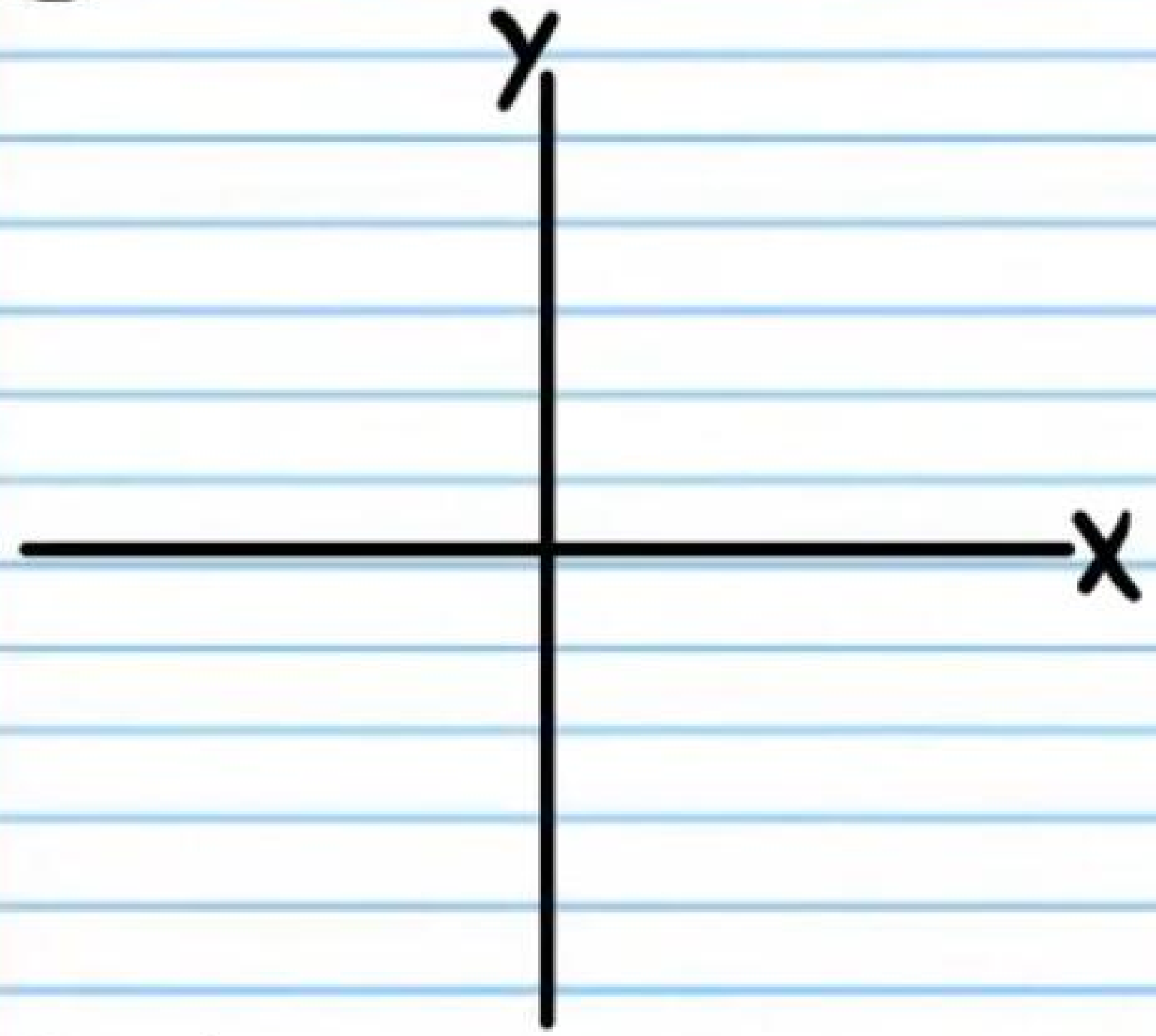
iv) $\csc\theta$

v) $\sec\theta$

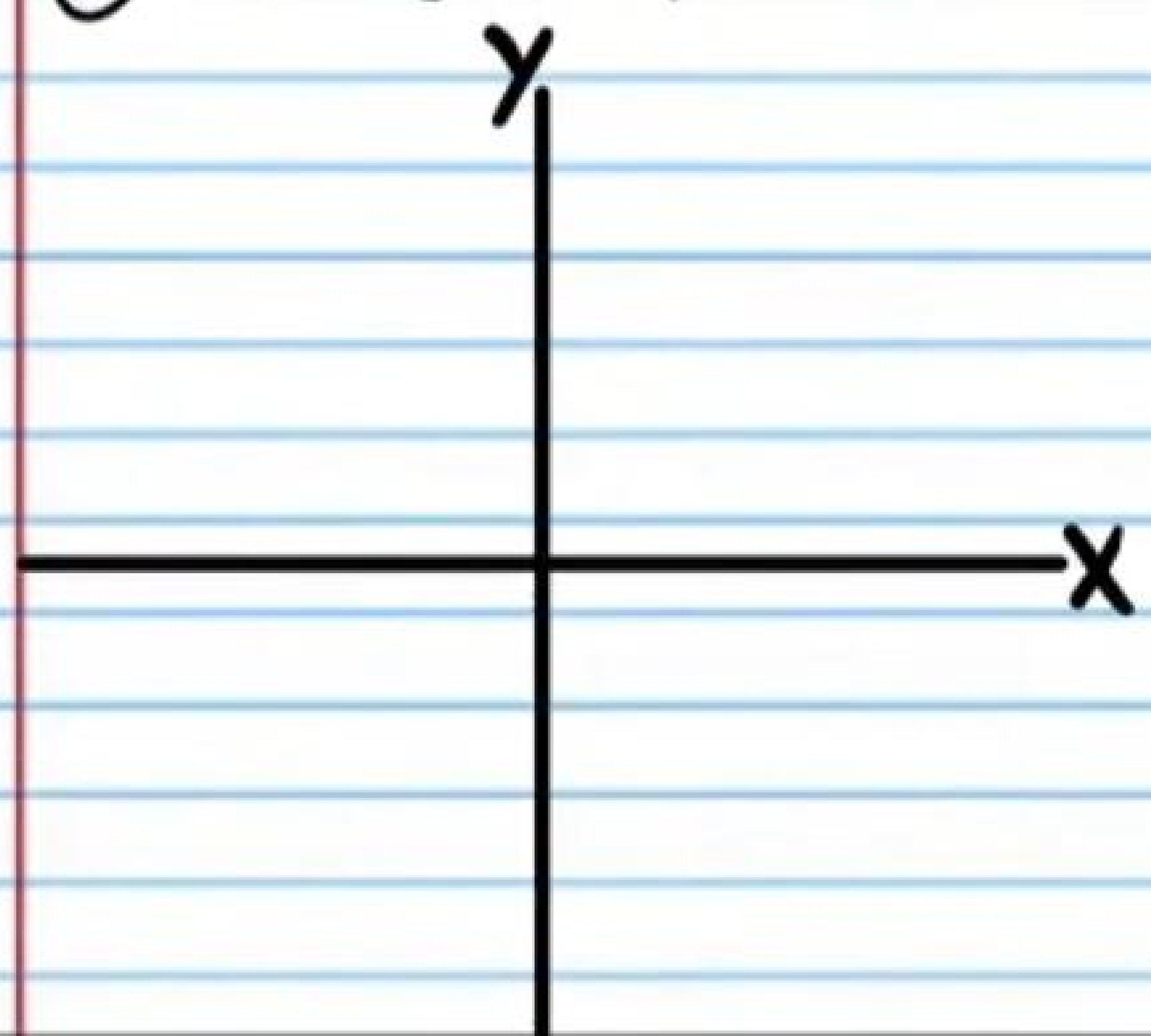
vi) $\cot\theta$

Find the values of the following expressions without a calculator. You must write exact answers. Draw each angle in standard position. Write the measure of each angle and the measure of its reference angle, along with the corresponding right triangle. If a triangle is collapsed, then you cannot draw it. In this scenario, draw only the angle.

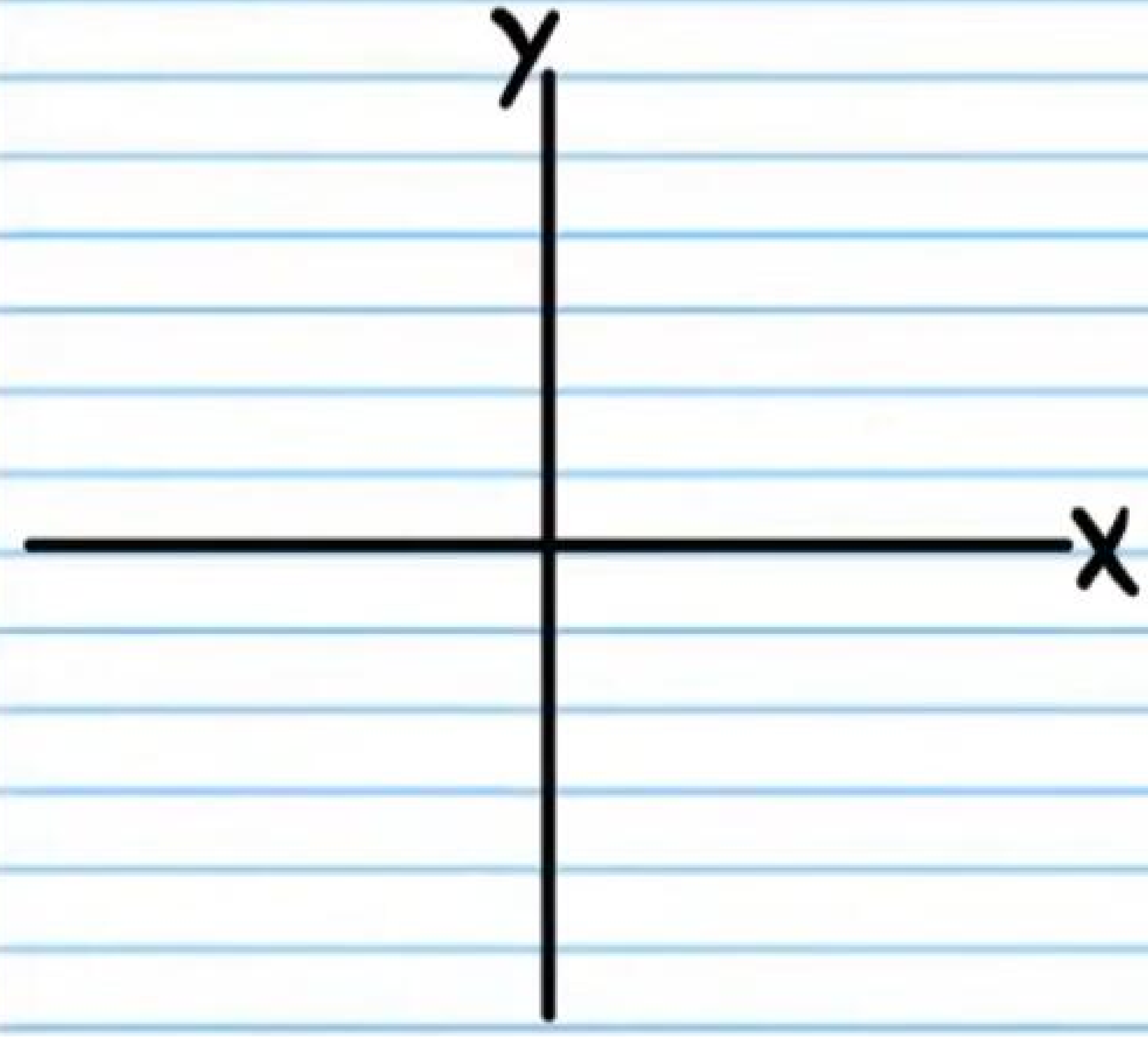
③ $\sec 240°$



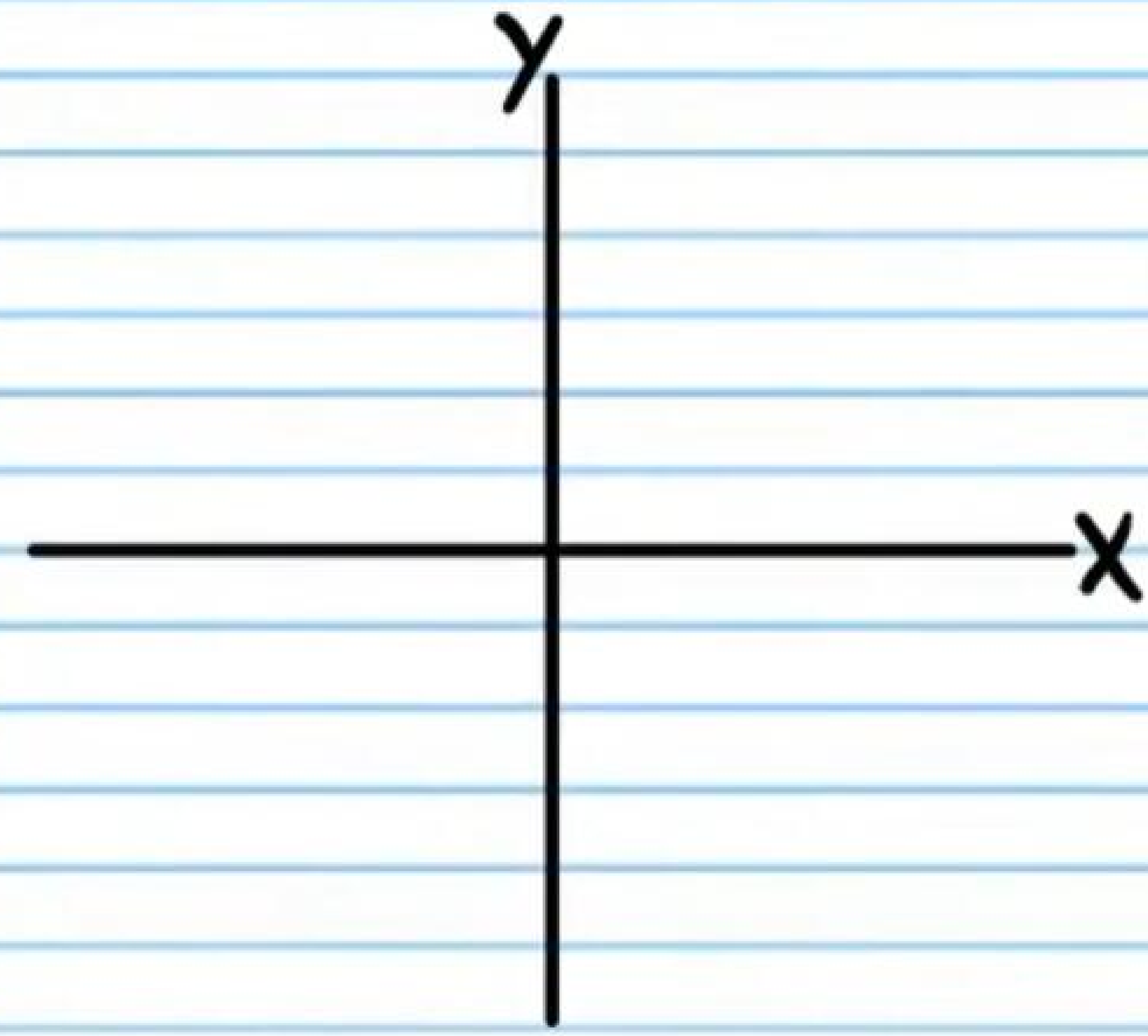
④ $\sin(-225°)$



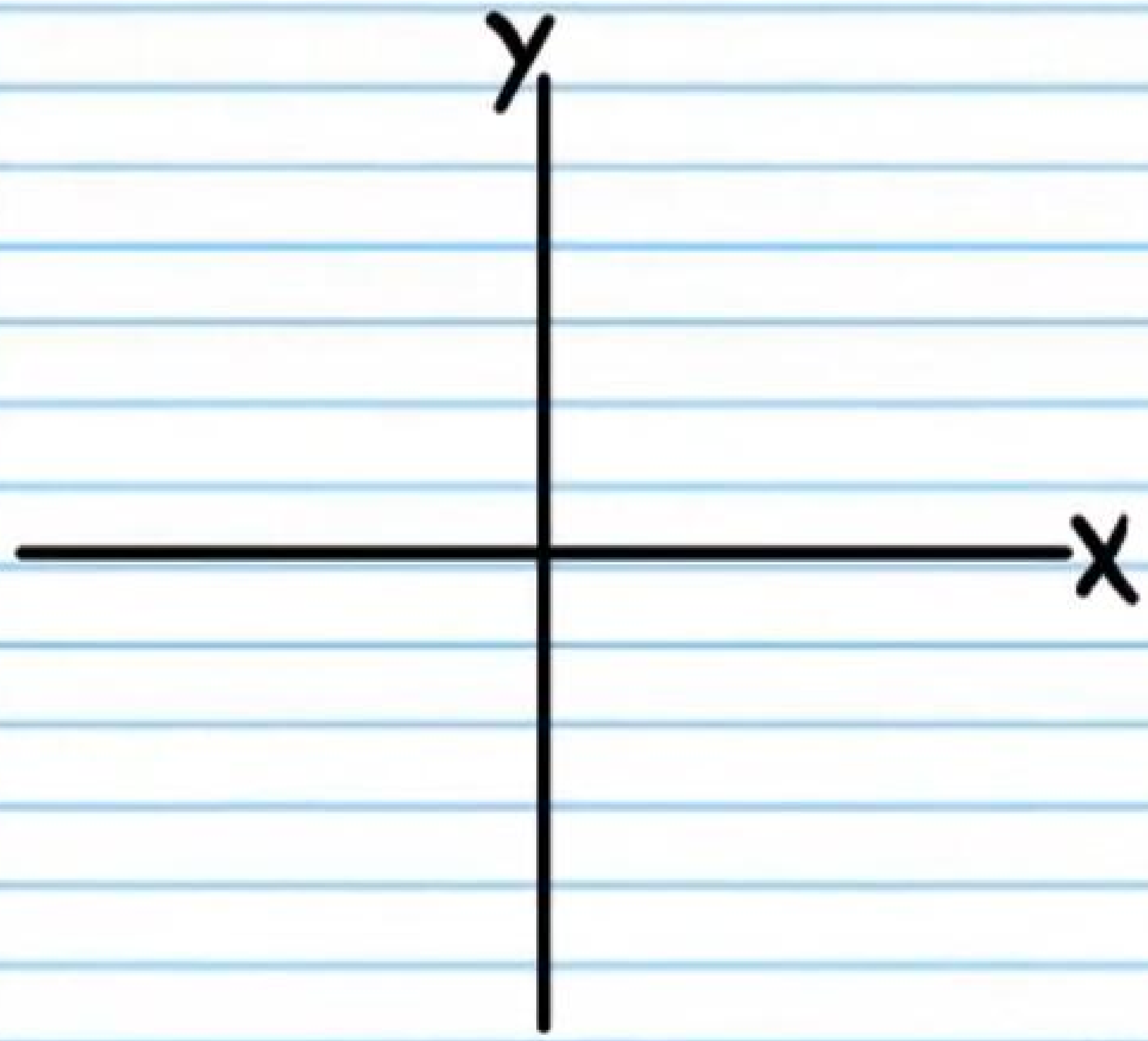
⑤ $\cos 180°$



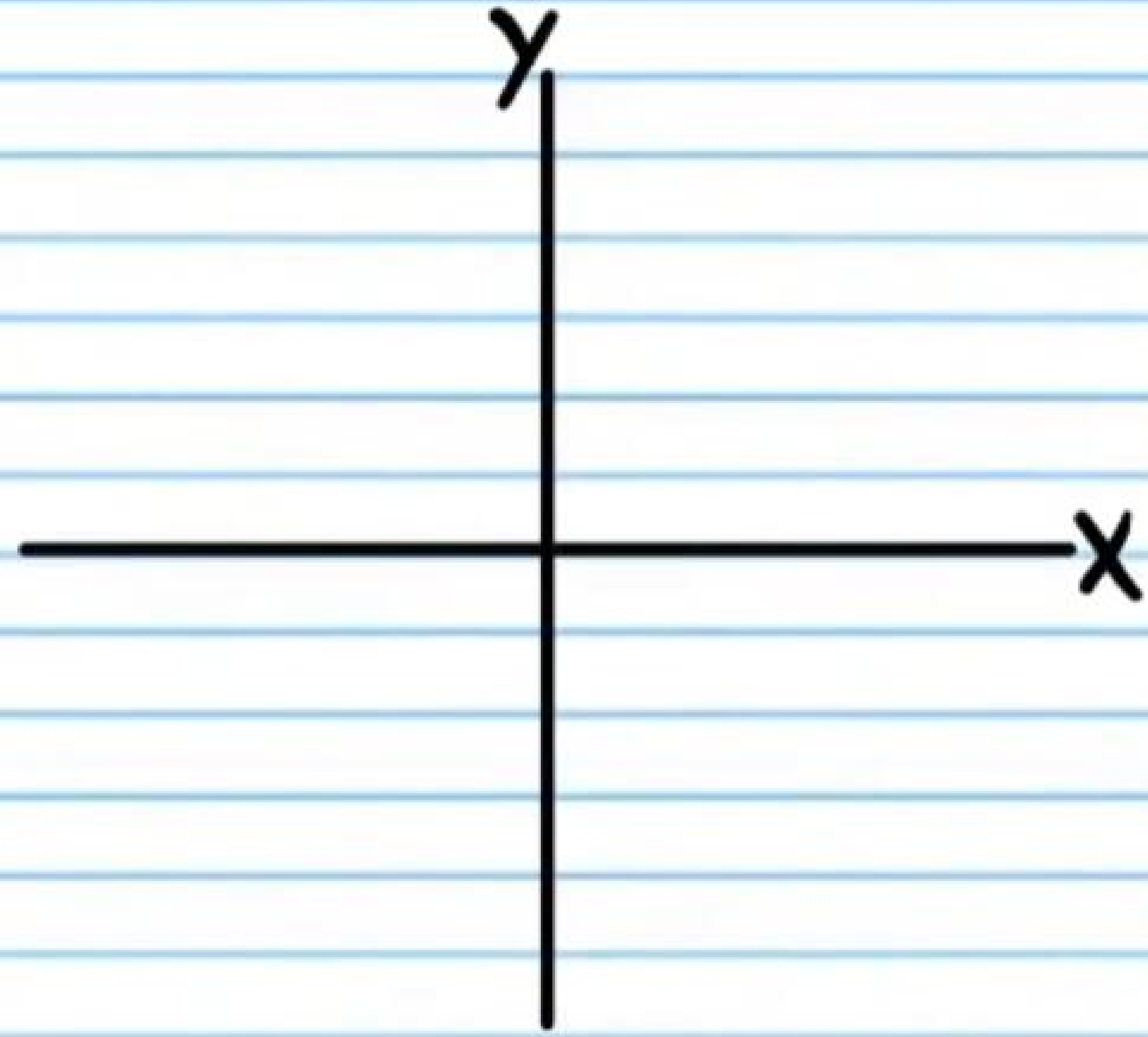
⑥ $\csc 30°$



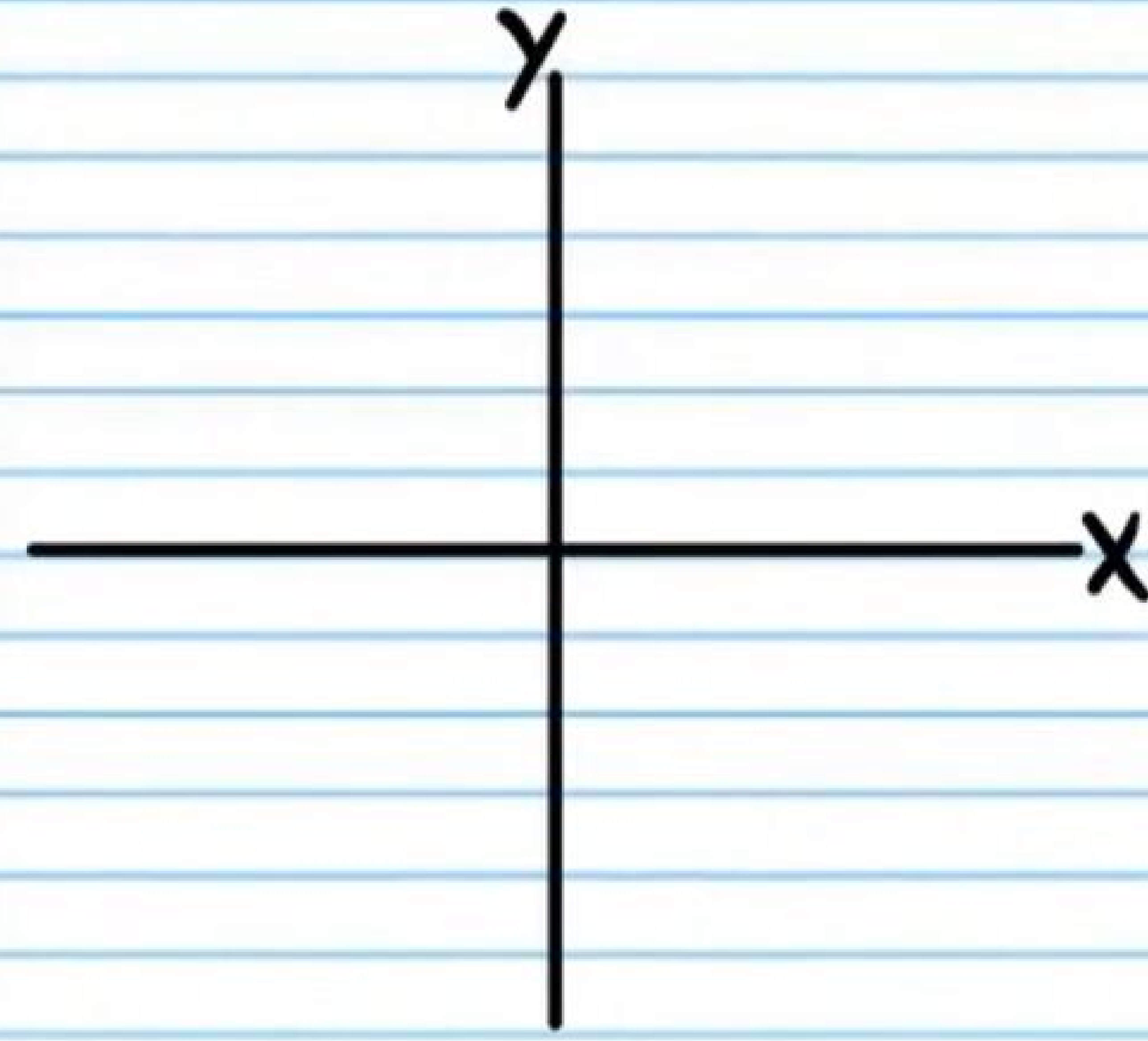
⑦ $\cos 90°$



⑧ $\cot(-60°)$

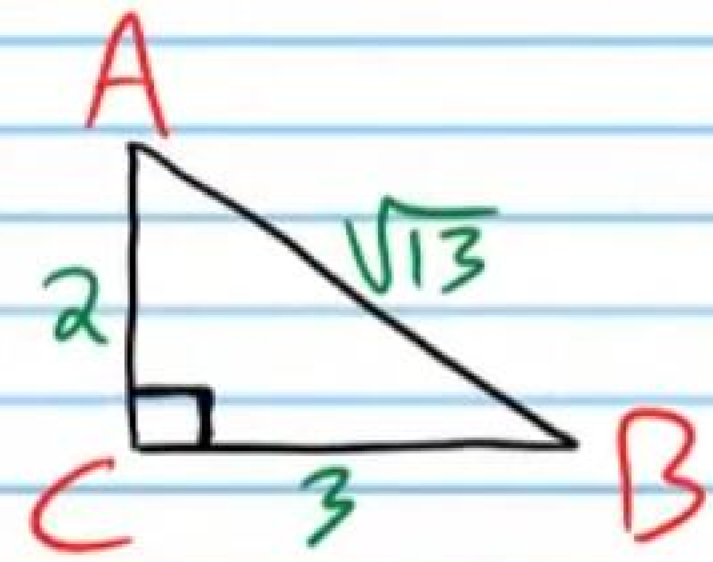


⑨ $\tan 270°$

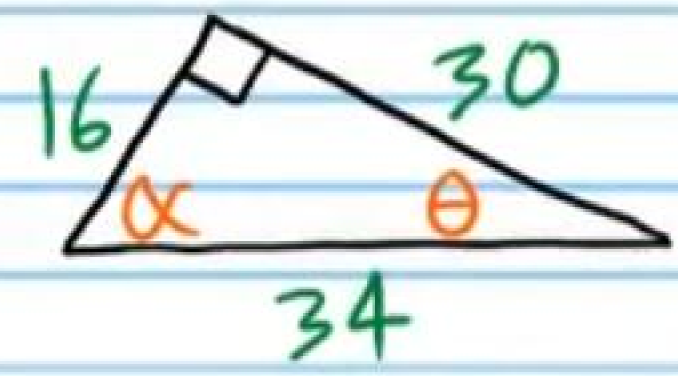


⑩ Use the triangles below to find the values of the expressions. If necessary, rationalize denominators.

i) $\tan A$

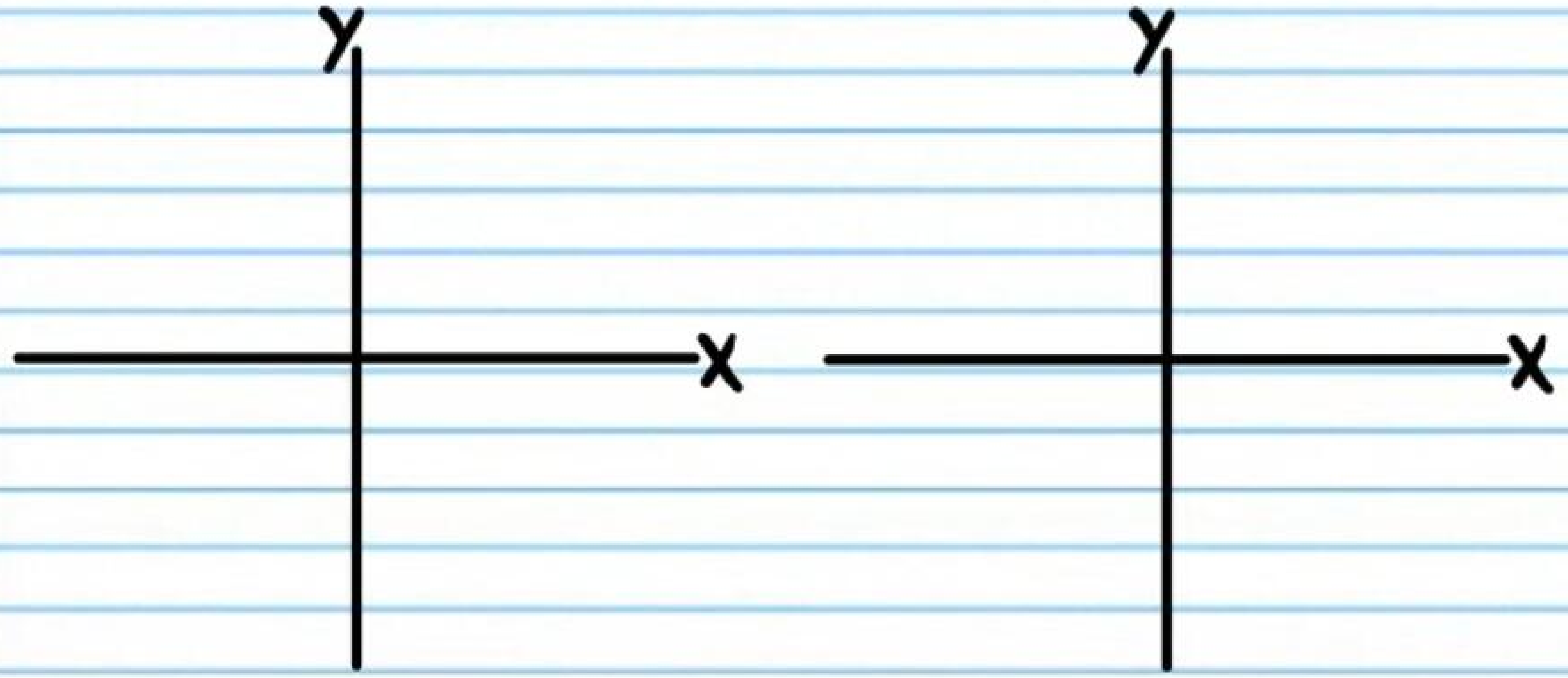


ii) $\csc\theta$

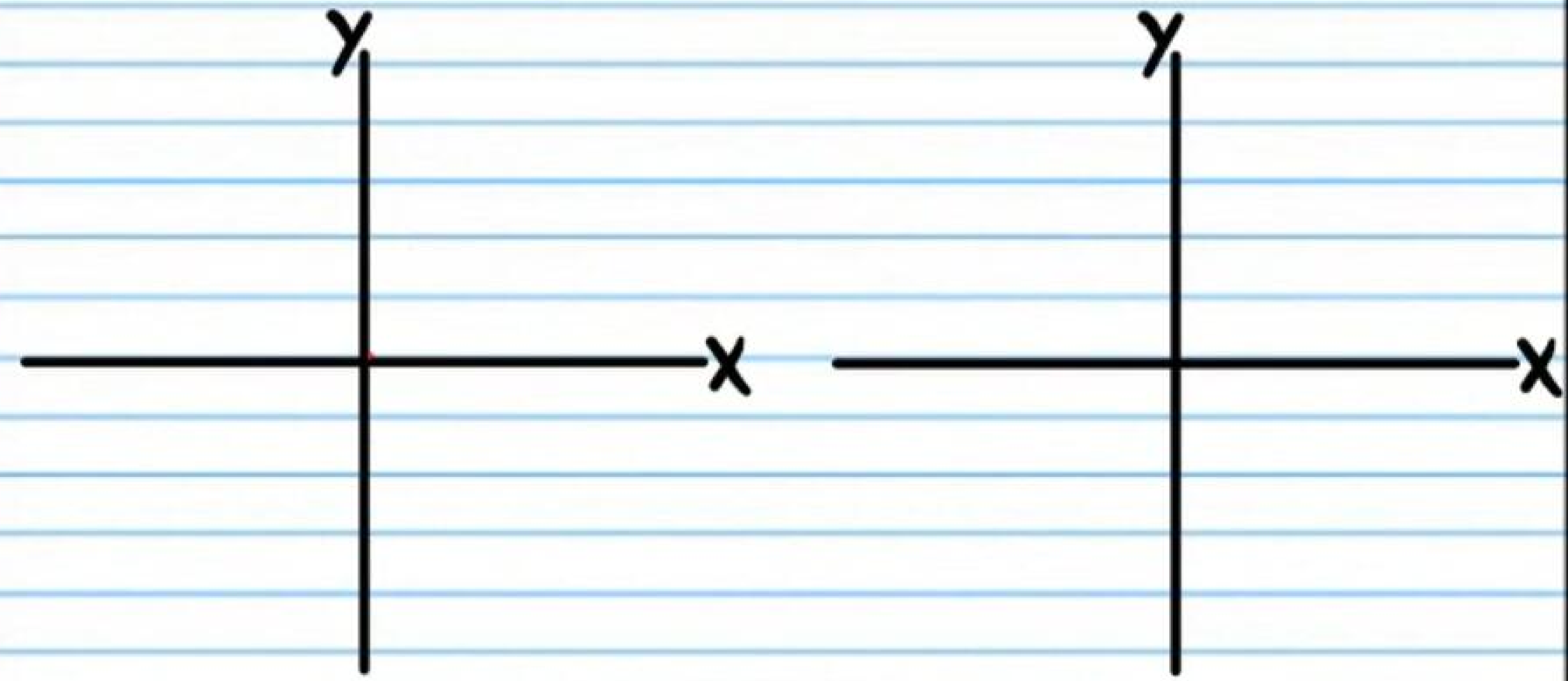


For problem #'s 11-15, draw the angles that are solutions to each equation in standard position as well as their corresponding right triangles. Write all three interior angle measures of each triangle and the relative side lengths.
Do not use a calculator.

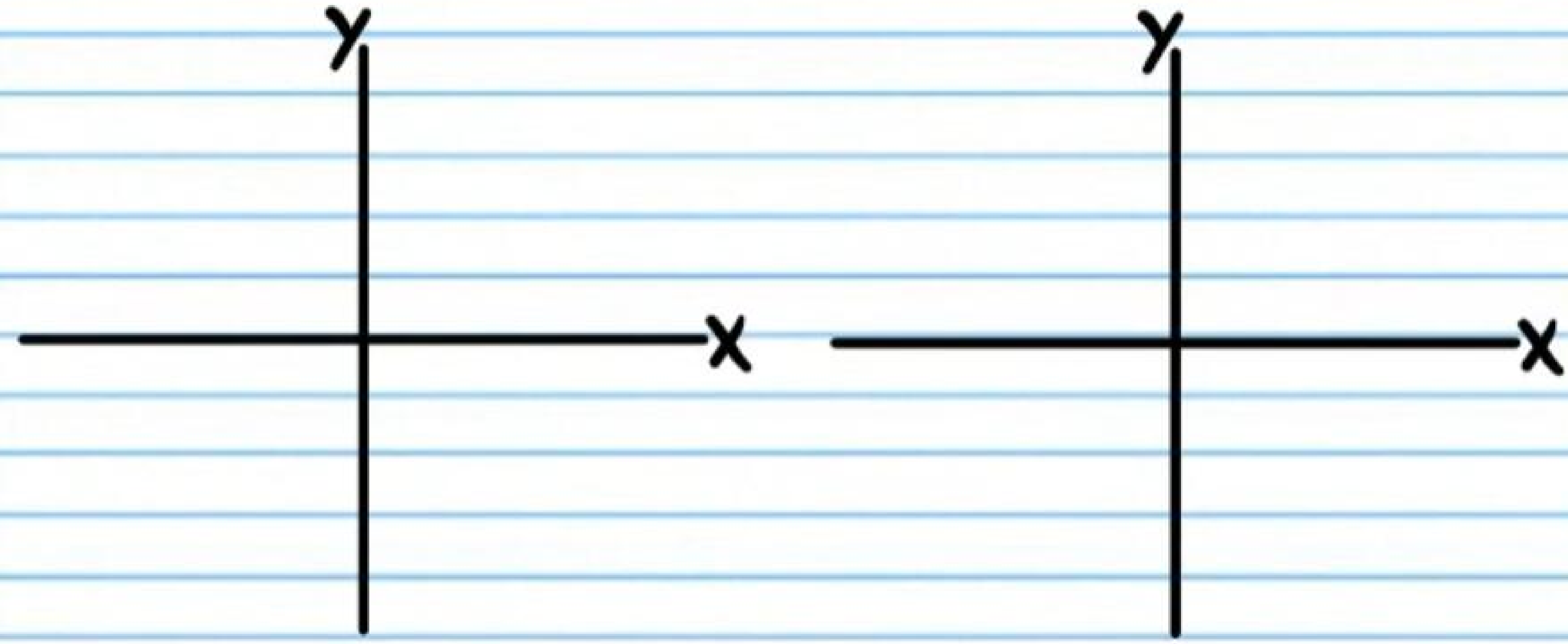
(11) If $\csc\theta = \sqrt{2}$, and $0° < \theta \leq 360°$, Find the values of θ.



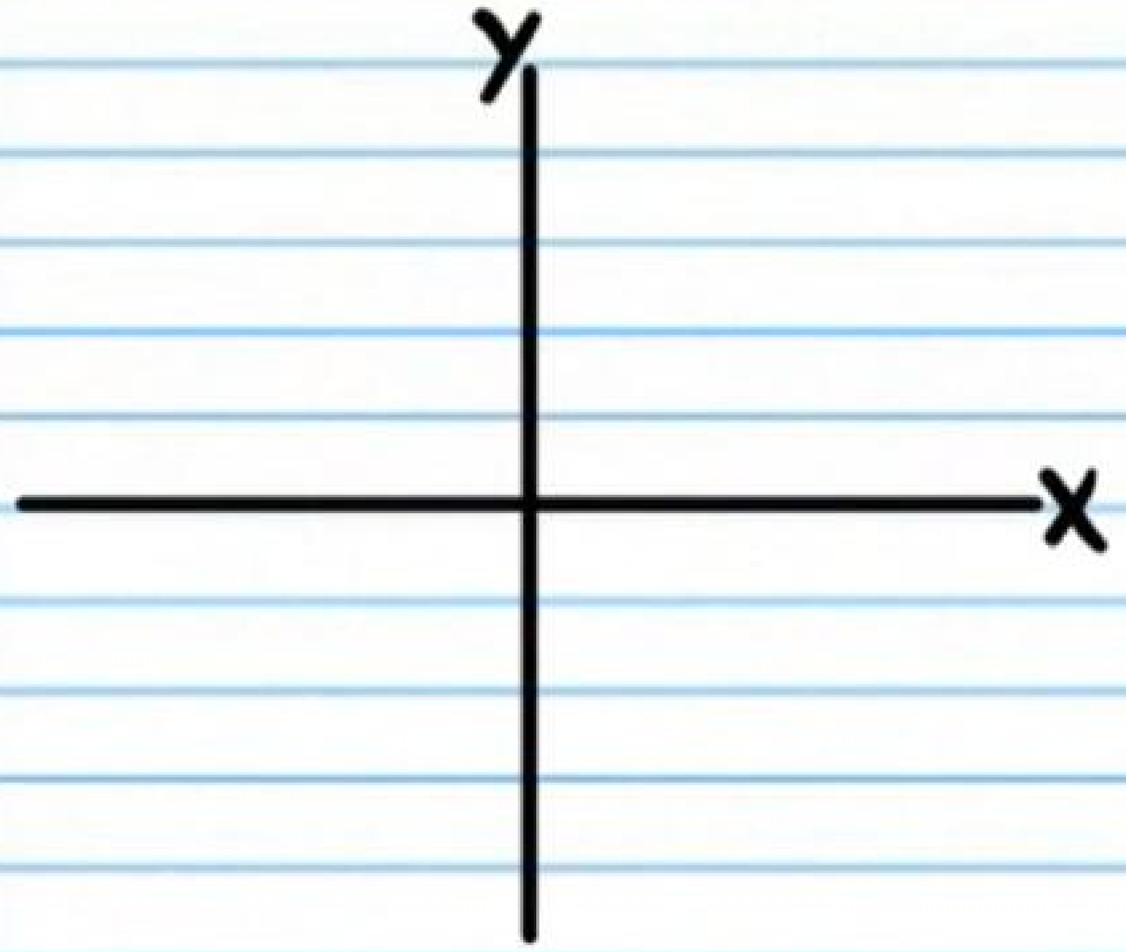
(12) If $\sin\theta = -\frac{\sqrt{3}}{2}$ and $0° < \theta \leq 360°$, Find the values of θ.

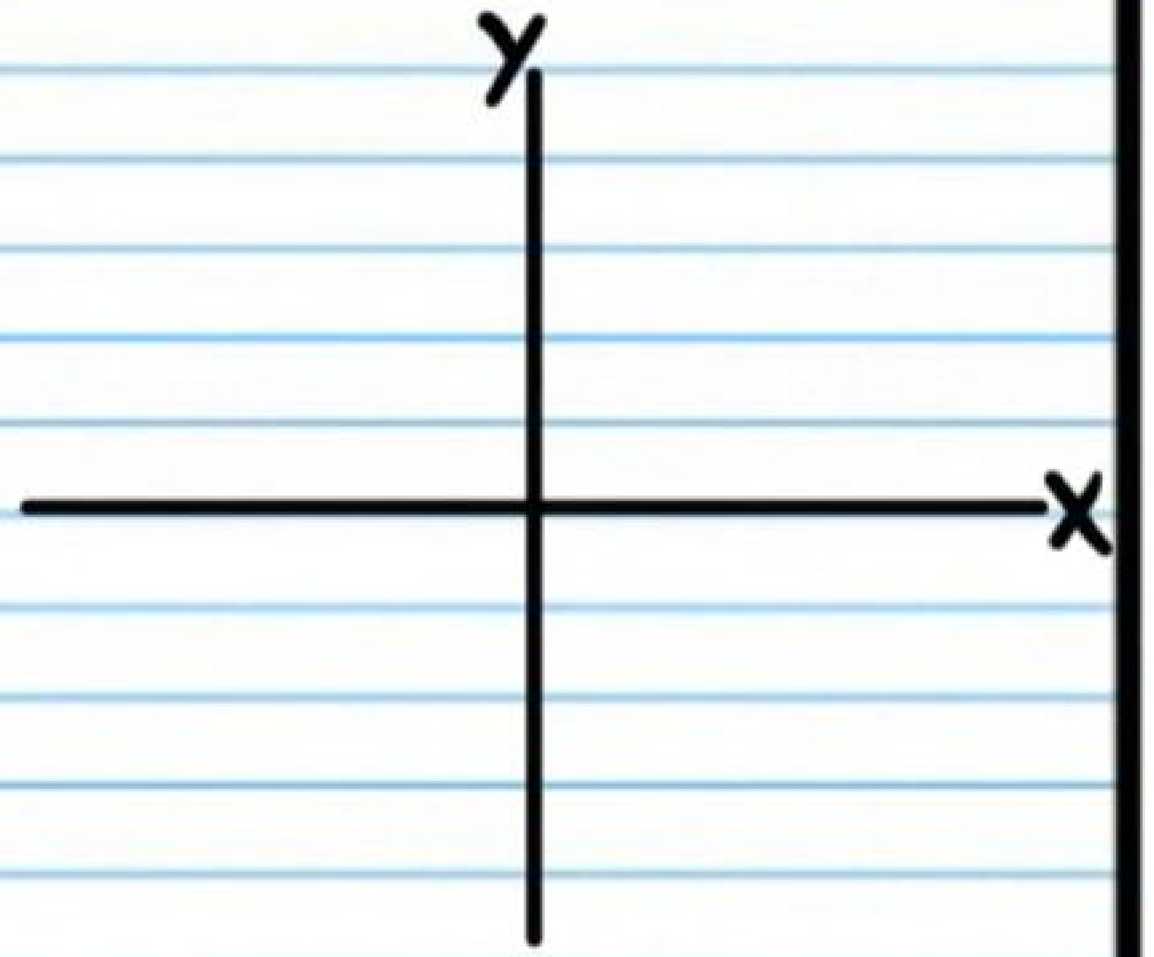


(13) If $\cot\theta = -1$, and $0° < \theta \leq 360°$, Find the values of θ.

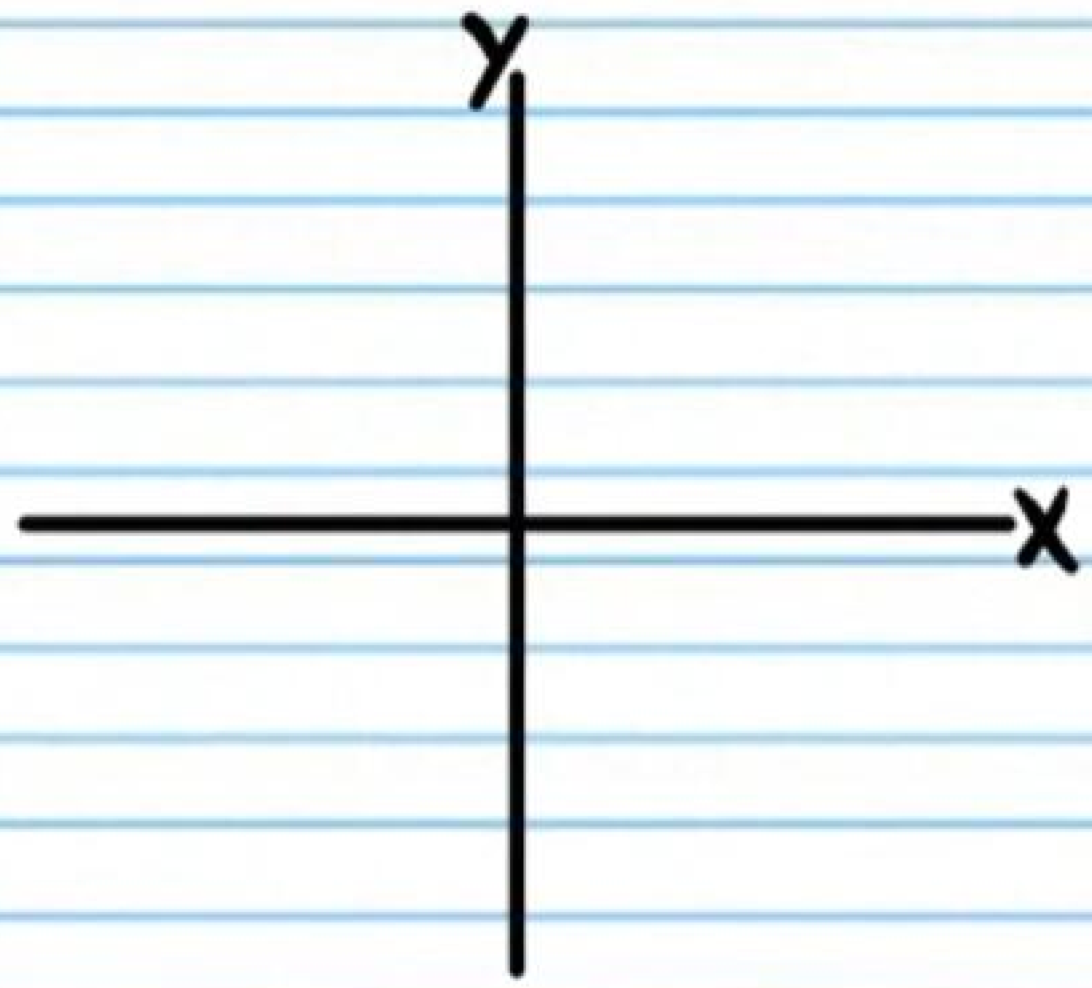


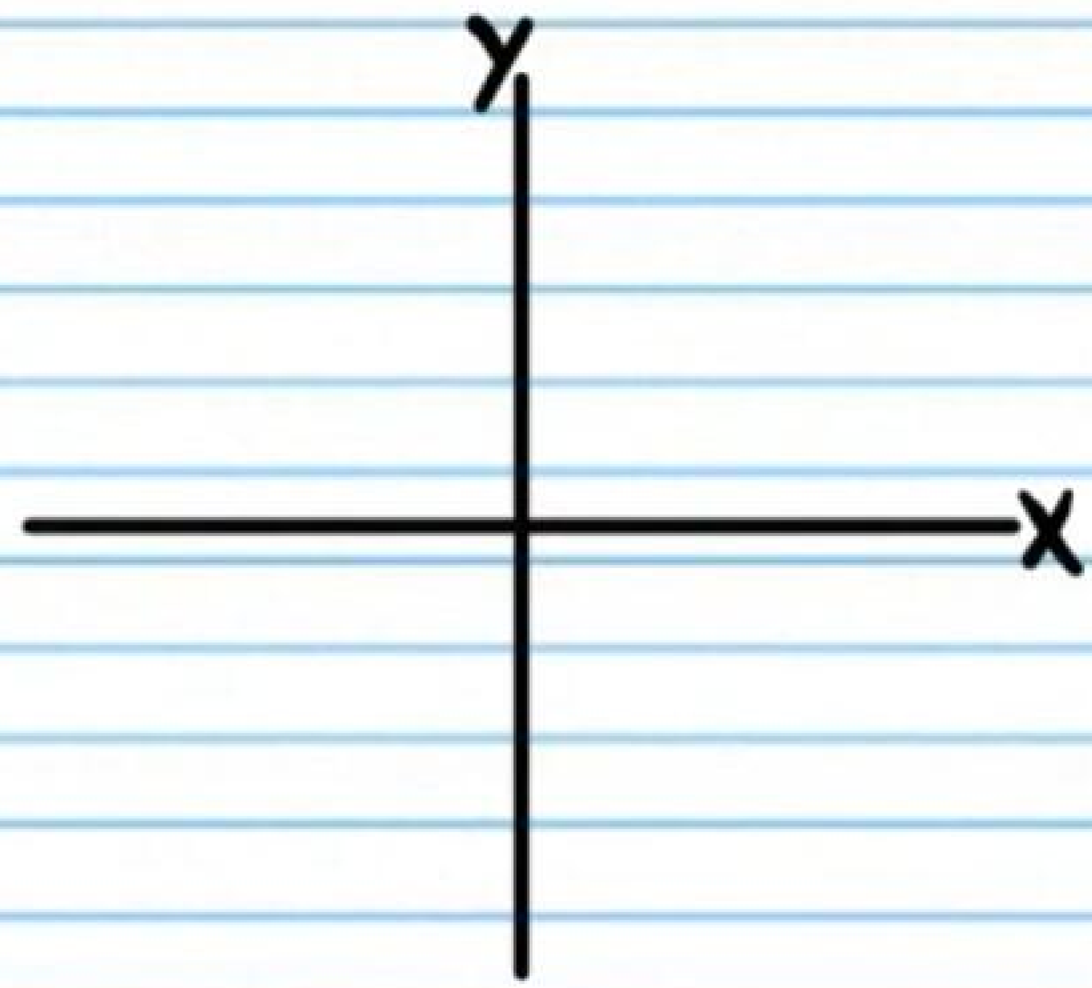
(14) If $\tan\theta = \frac{1}{\sqrt{3}}$, find all values of θ.





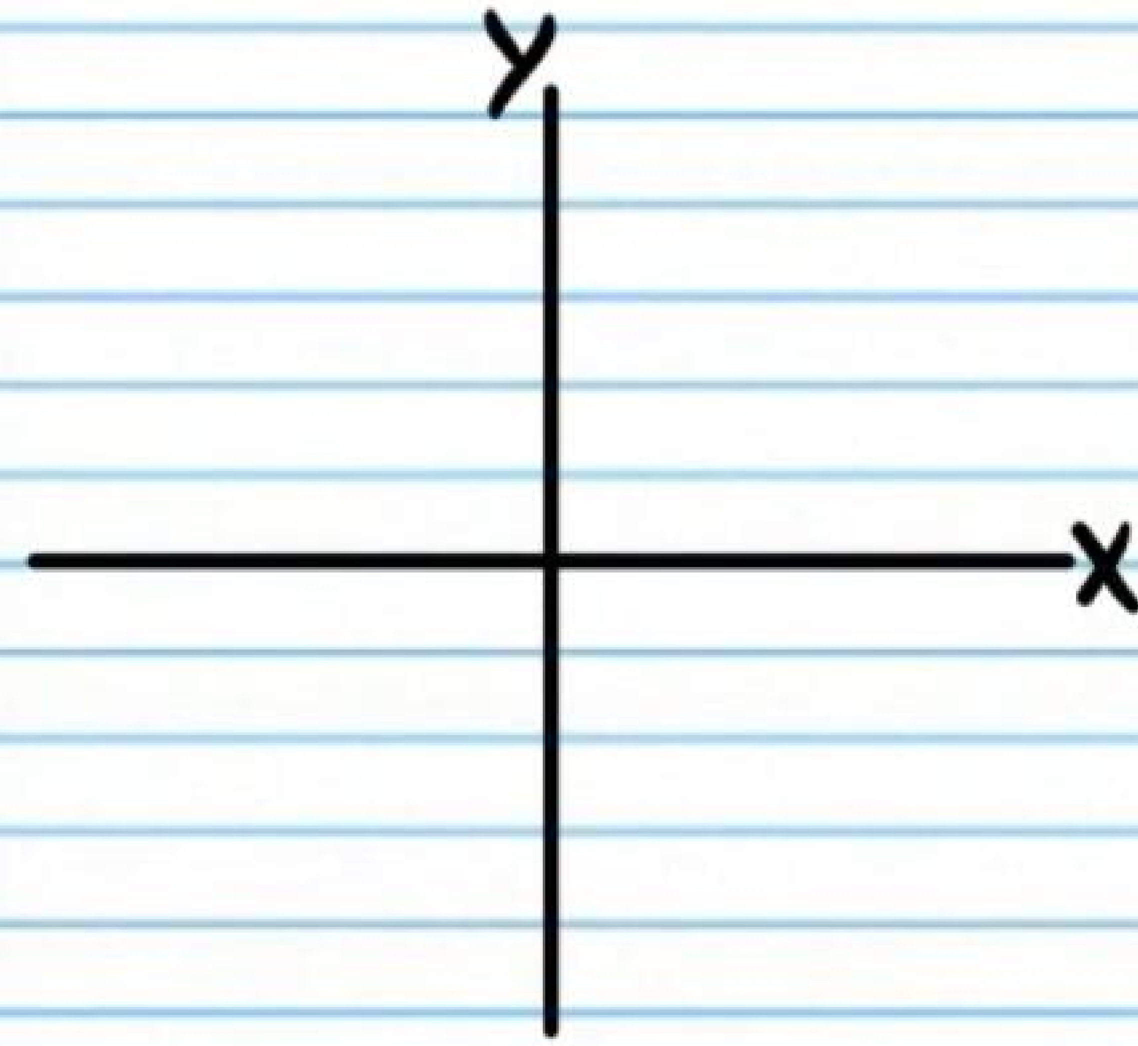
(15) If $\sec\theta = \sqrt{2}$, find all values of θ.

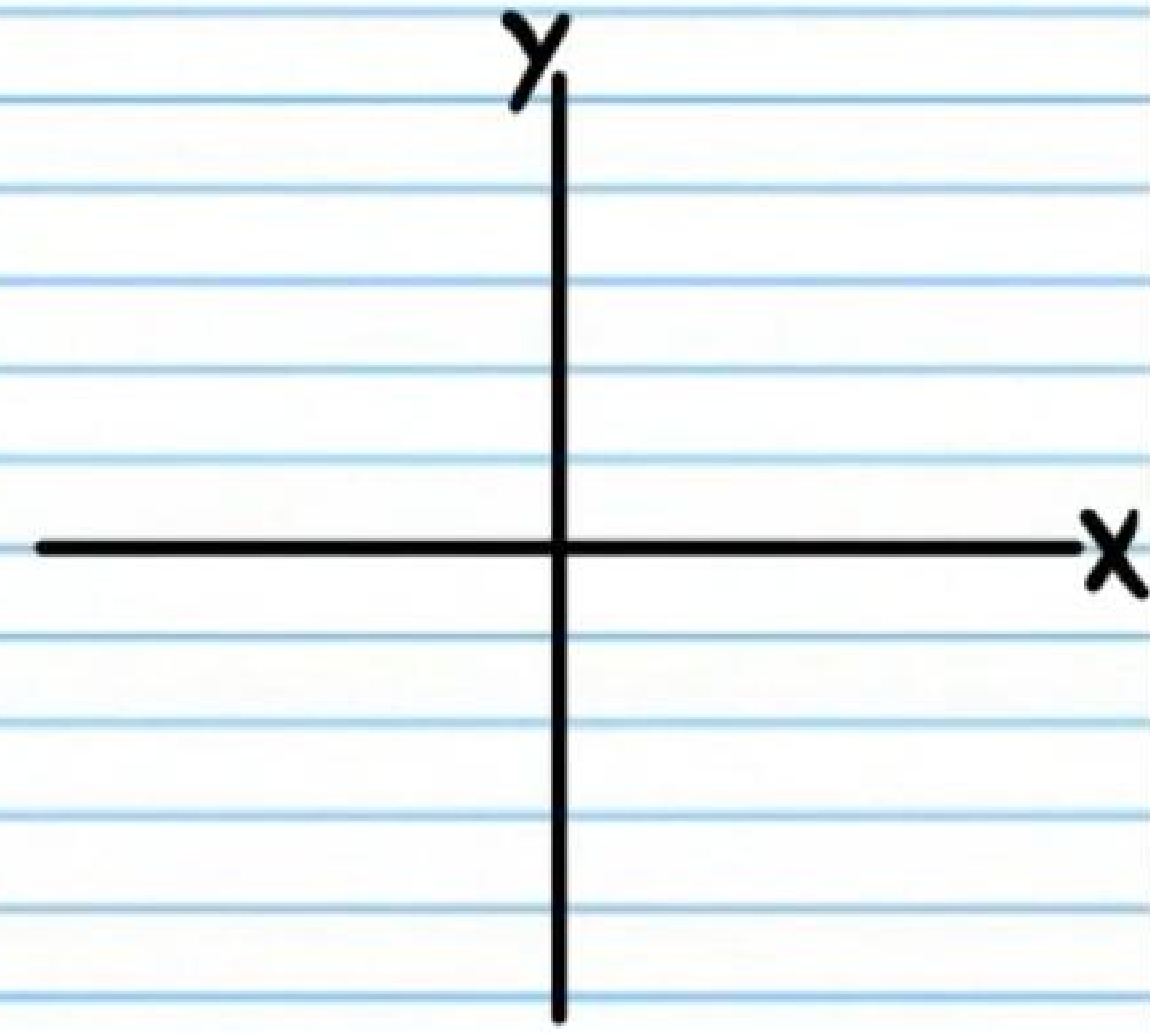




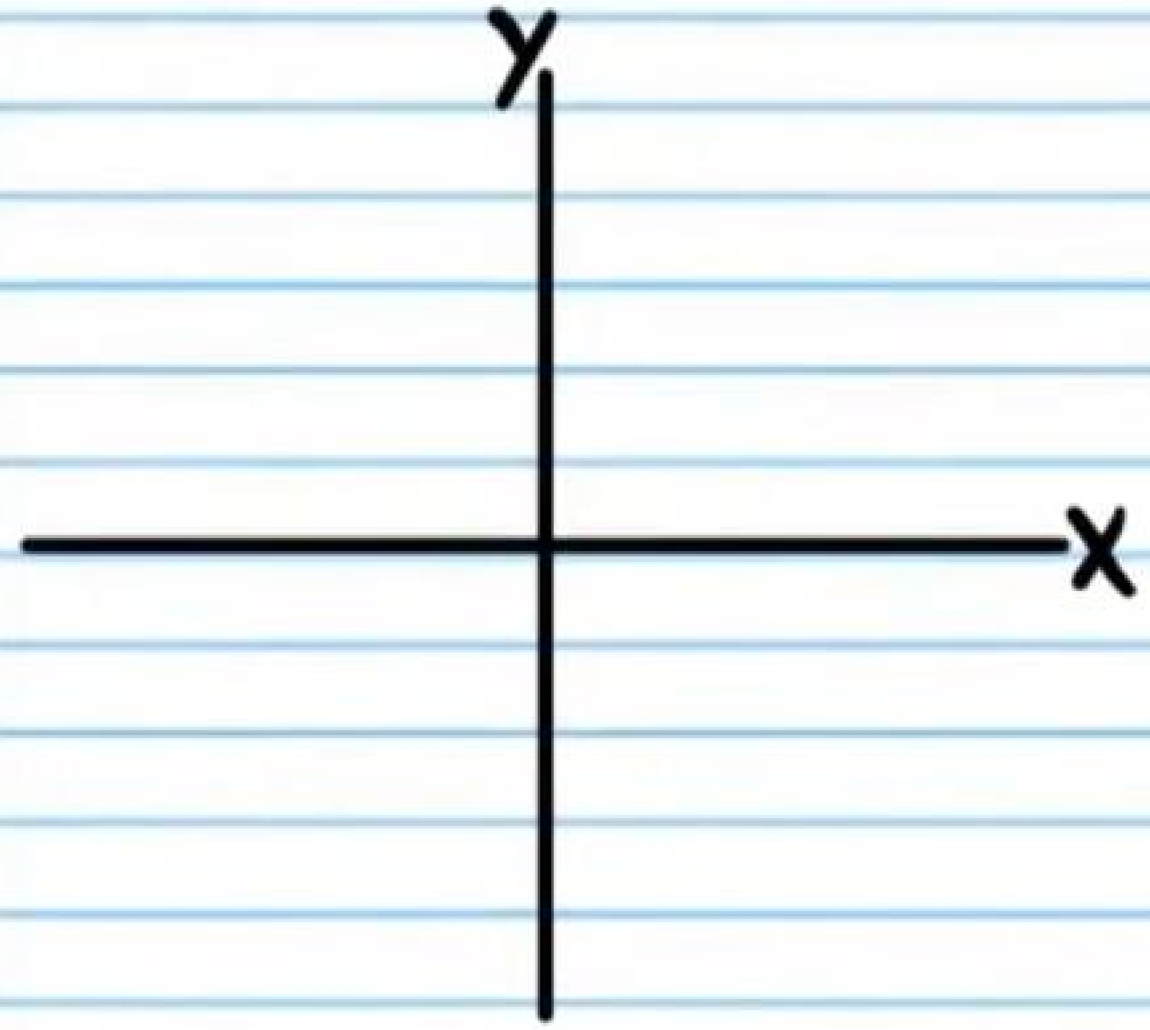
For problem #'s 16-18, use a calculator to find the degree measures. Draw each angle in standard position, the corresponding right triangle, and the reference angle.

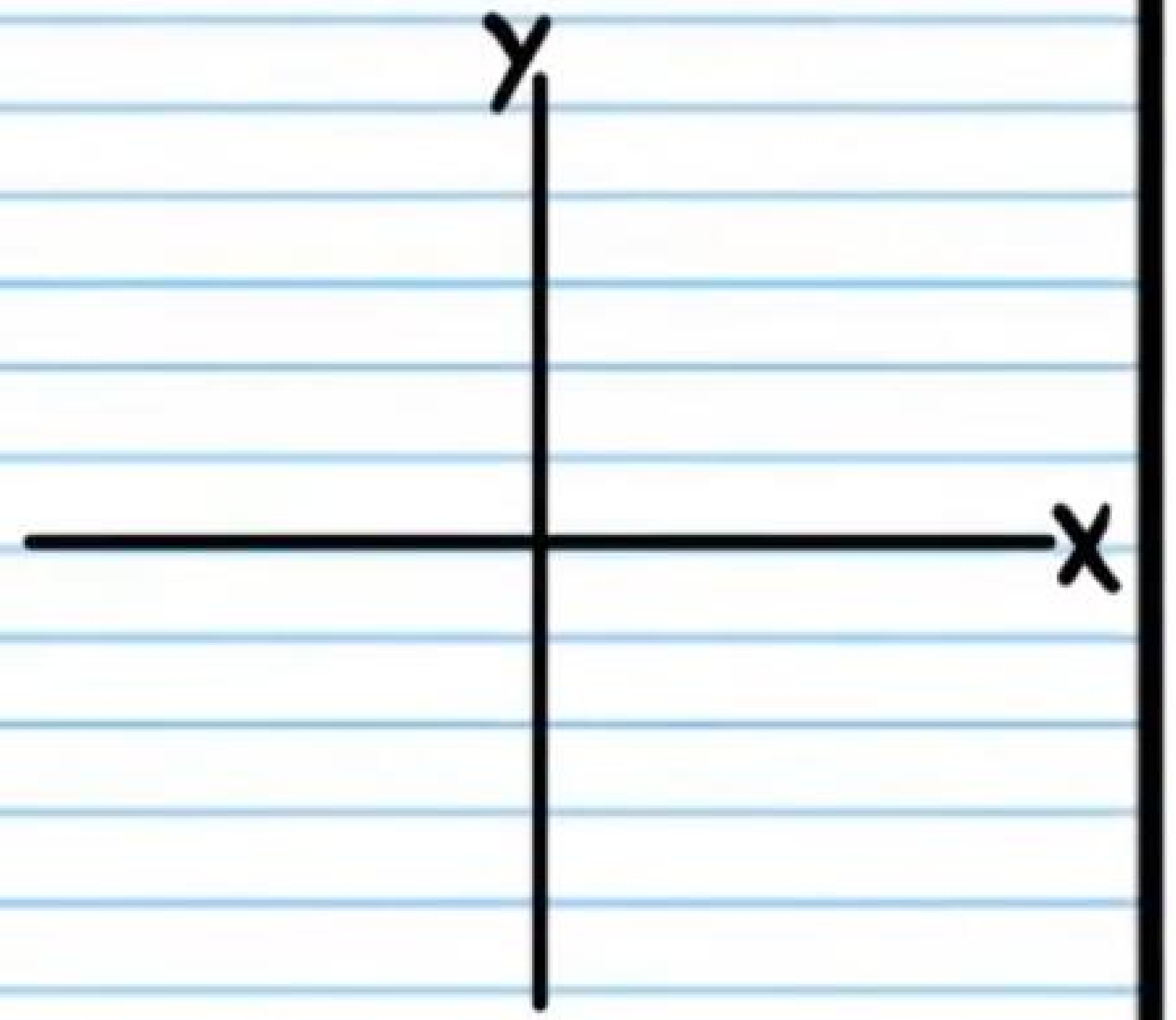
(16) If $\cos\theta = -\frac{1}{6}$, and $0° < \theta \leq 360°$, Find the values of θ.



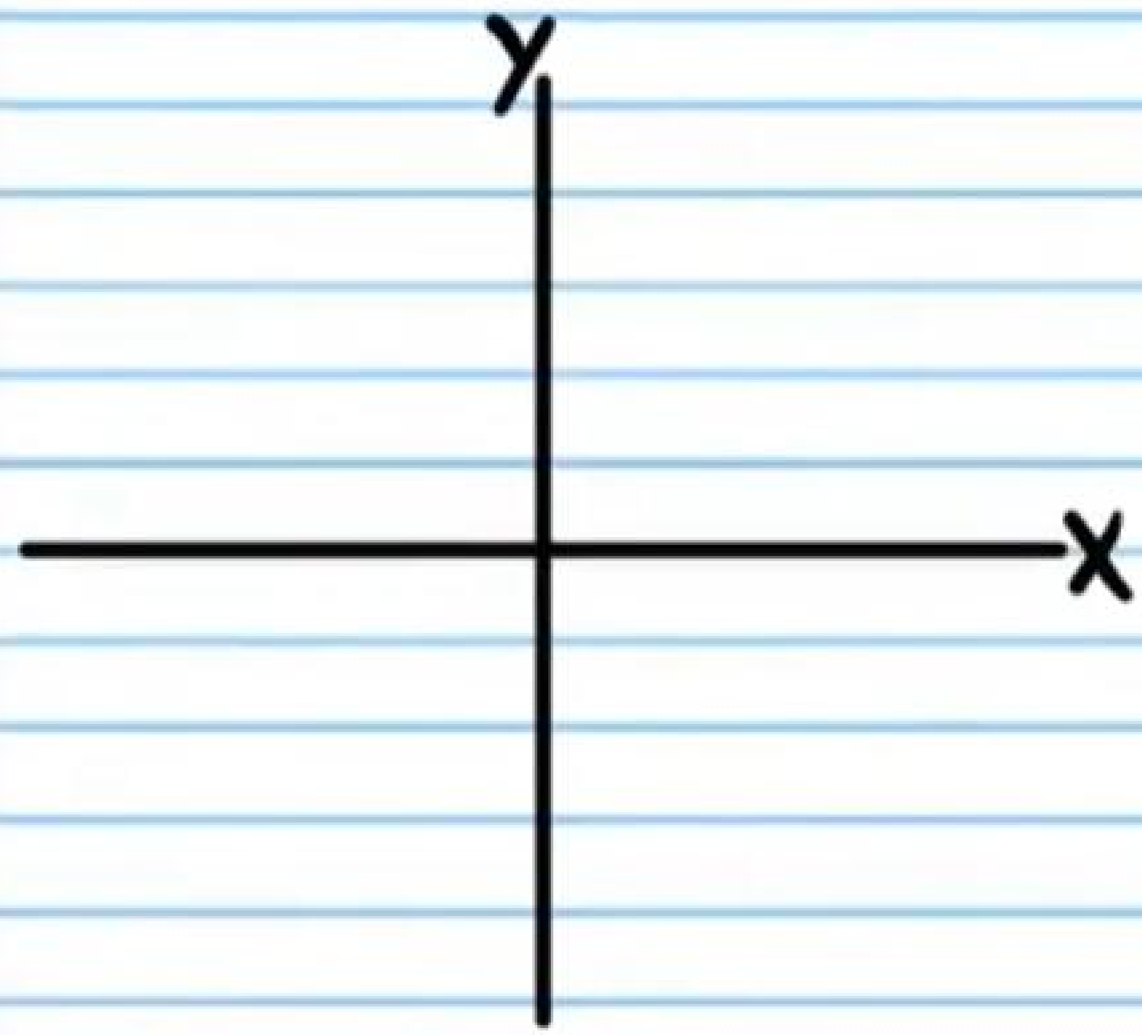


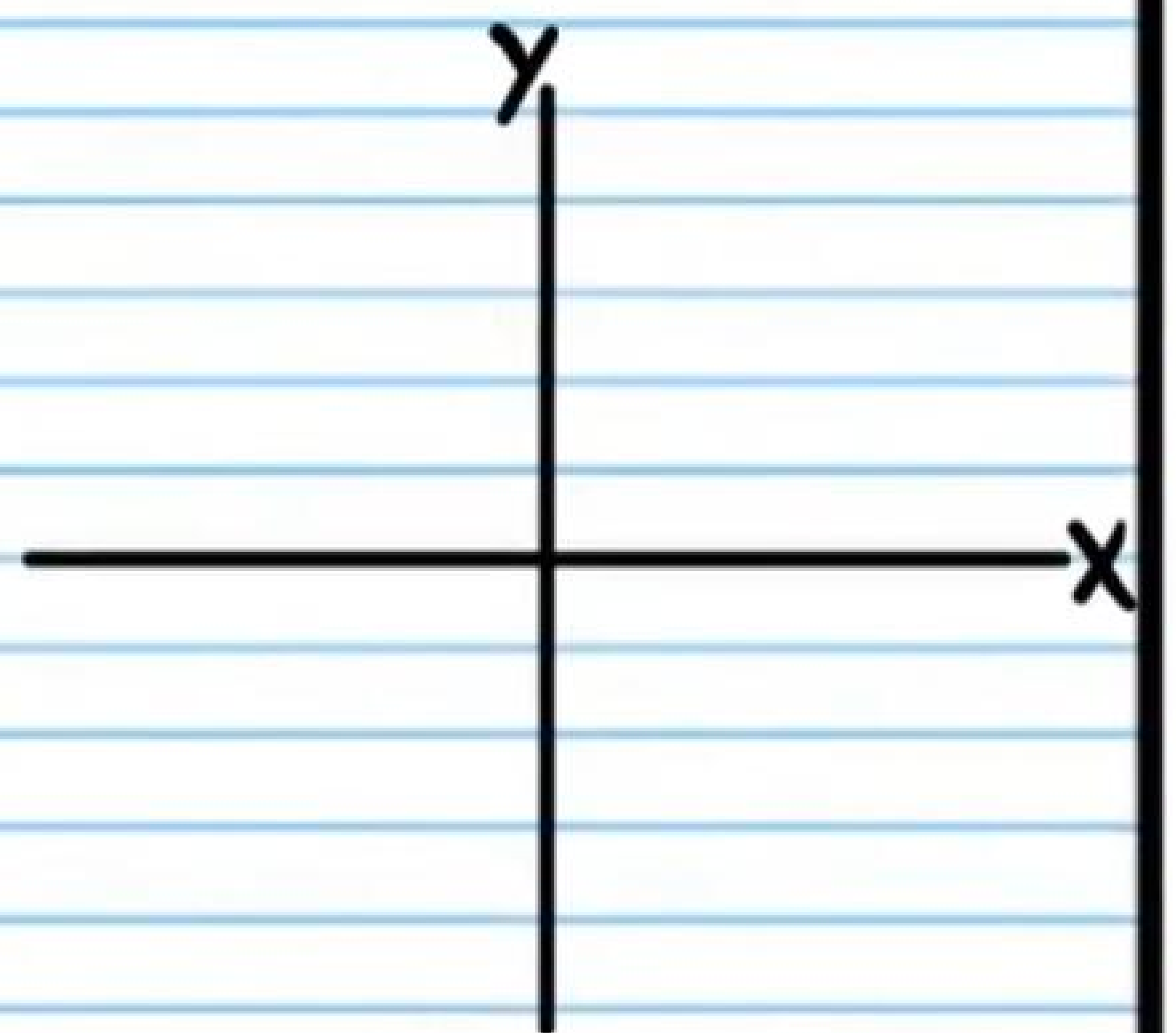
⑰ If $\sin\theta = -0.89$, and $0° < \theta \leq 360°$, Find the values of θ.



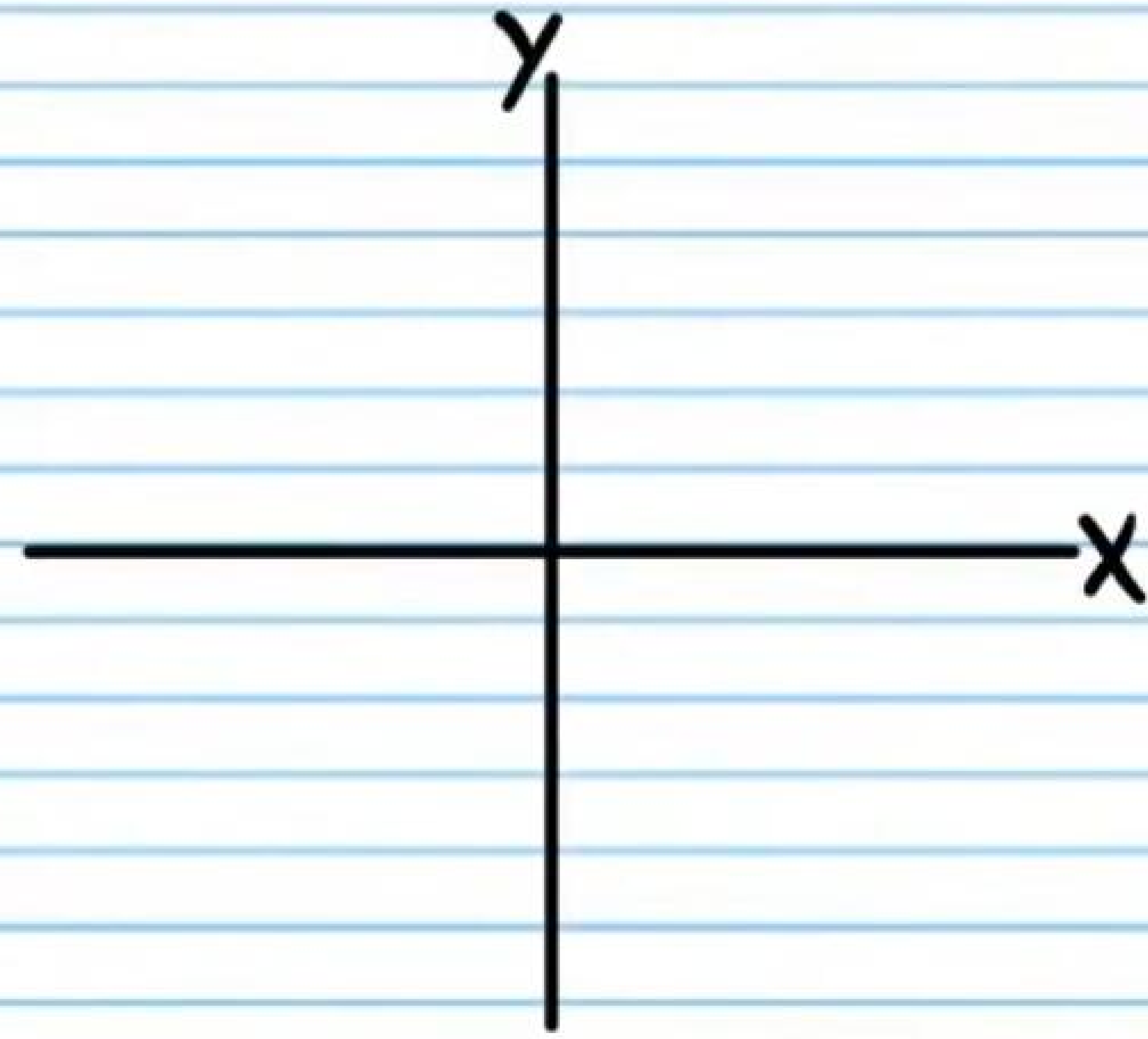


⑱ If $\tan\theta = 0.4$, and $0° < \theta \leq 360°$, Find the values of θ.





⑲ If $\sec\beta = -\frac{7}{5}$, and β terminates in quadrant II, find the values of the five remaining basic trigonometric functions of β.



i) $\cot\beta$

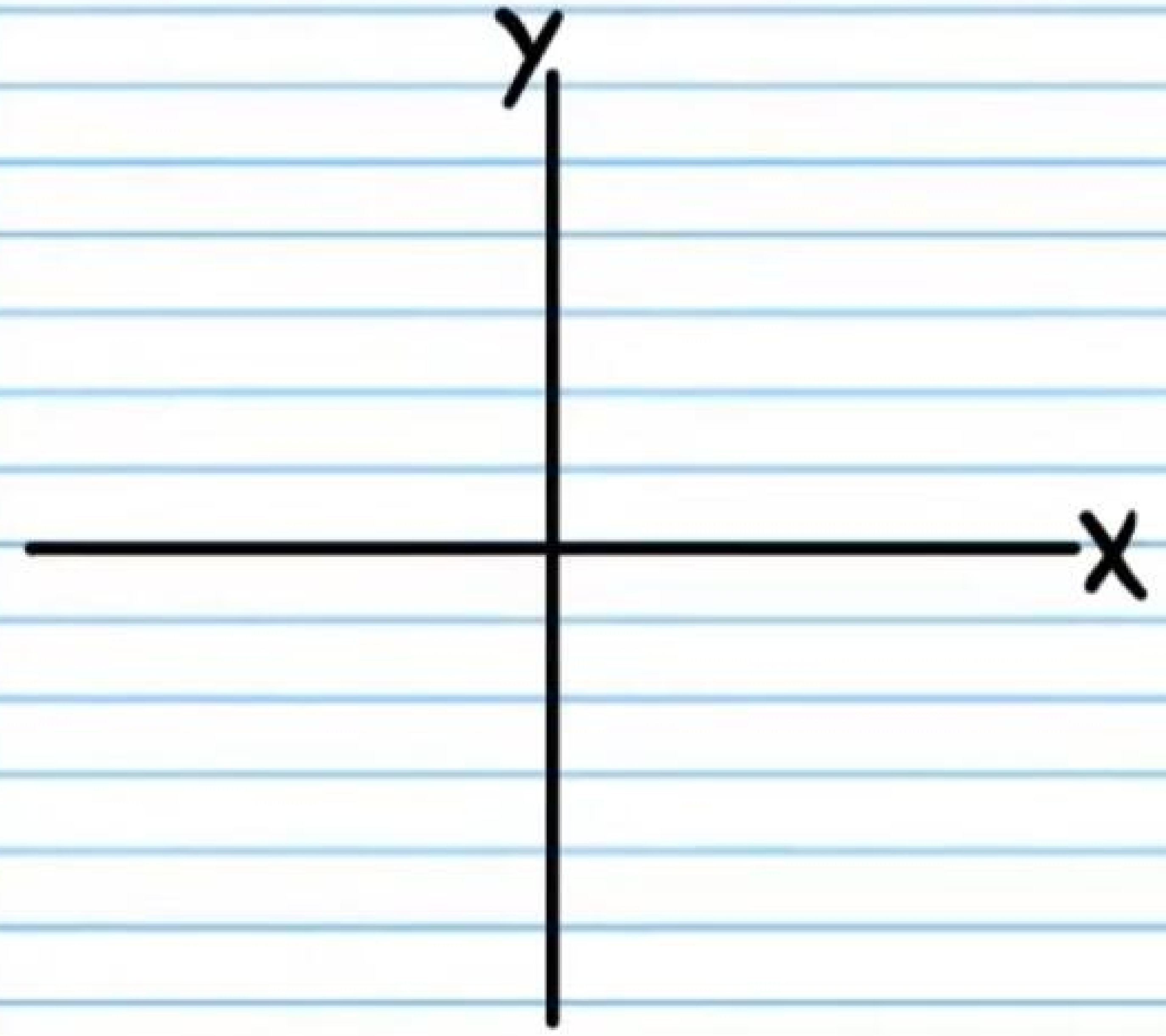
ii) $\cos\beta$

iii) $\sin\beta$

iv) $\csc\beta$

v) $\tan\beta$

(20) If $\csc\theta = -\frac{4}{3}$ and θ terminates in quadrant III, find the values of the five remaining basic trigonometric functions of θ.



i) $\csc\theta$

ii) $\sec\theta$

iii) $\cot\theta$

iv) $\cos\theta$

v) $\sin\theta$

(21) The measure of an angle in standard position is written below. Find the measures of three coterminal angles in standard position.

$59°$

θ_1

θ_2

θ_3

(22) If $\sin\theta = -1$, and $0 \leq \theta \leq 360°$, find the values of θ.

(23) If $\sec\theta = 1$, find all values of θ.

(24) If $\csc\theta = 1$, and $0 \leq \theta \leq 360°$, find the Values of θ.

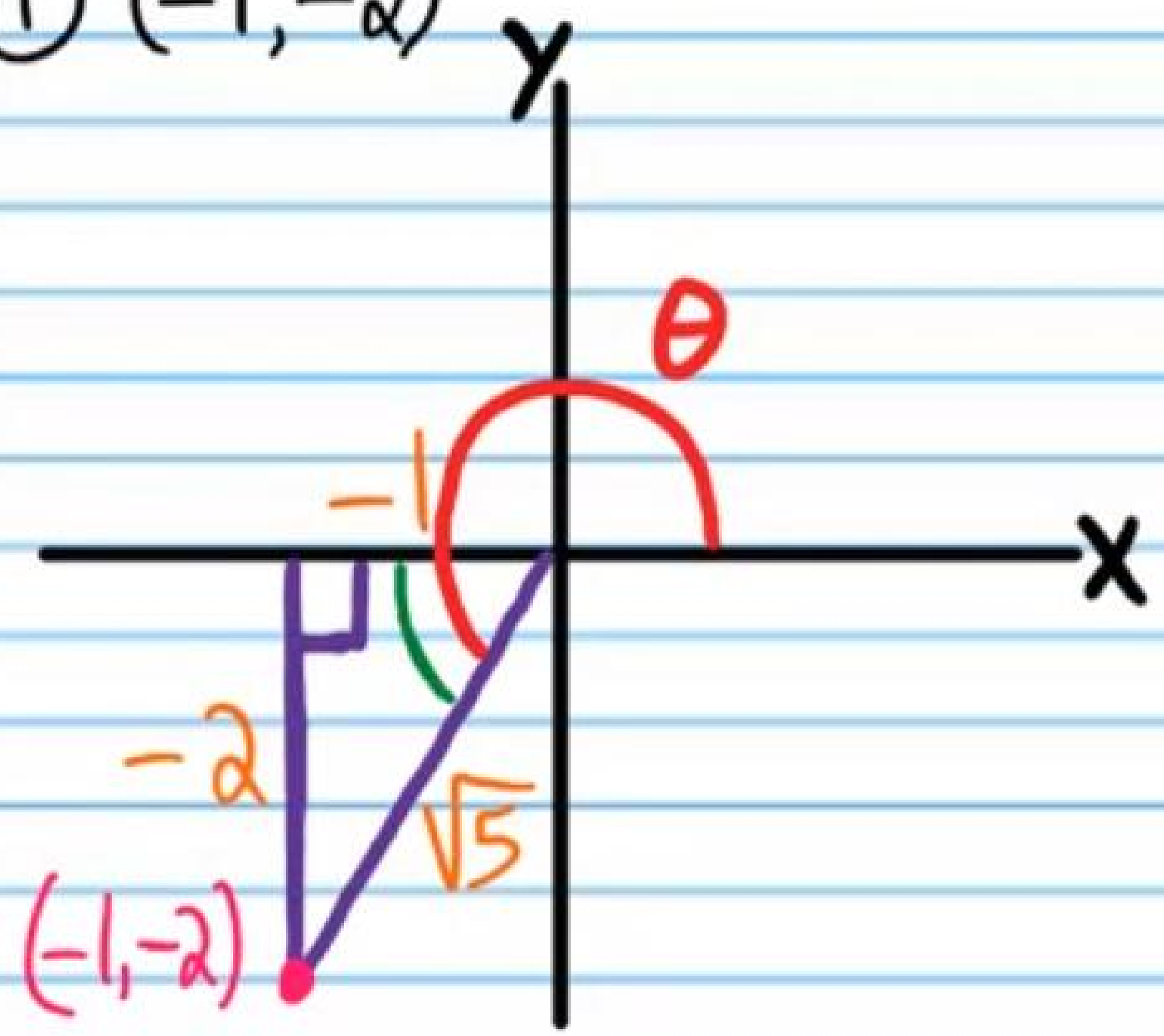
(25) Find the values of the expressions below with a calculator. Round to the nearest thousandth.

i) $\sin 329°$

ii) $\tan 100°$

Test I Answers

① $(-1,-2)$



i) $-\frac{2}{\sqrt{5}} = \boxed{-\frac{2\sqrt{5}}{5}}$

ii) $-\frac{1}{\sqrt{5}} = \boxed{-\frac{\sqrt{5}}{5}}$

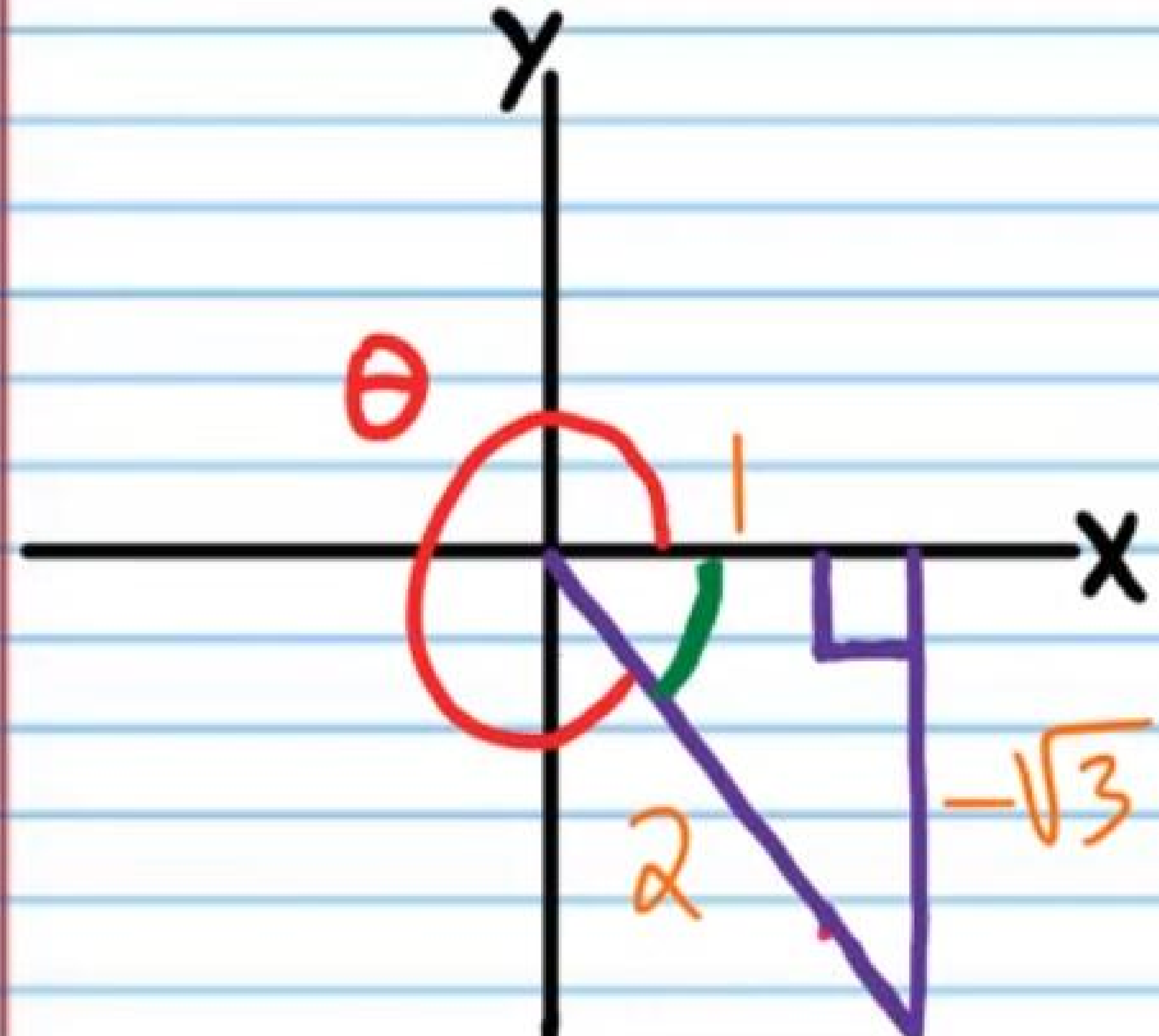
iii) $\boxed{2}$

iv) $\boxed{-\frac{\sqrt{5}}{2}}$

v) $\boxed{-\sqrt{5}}$

vi) $\boxed{\frac{1}{2}}$

② $(1,-\sqrt{3})$



i) $\sin\theta = \boxed{-\frac{\sqrt{3}}{2}}$

ii) $\cos\theta = \boxed{\frac{1}{2}}$

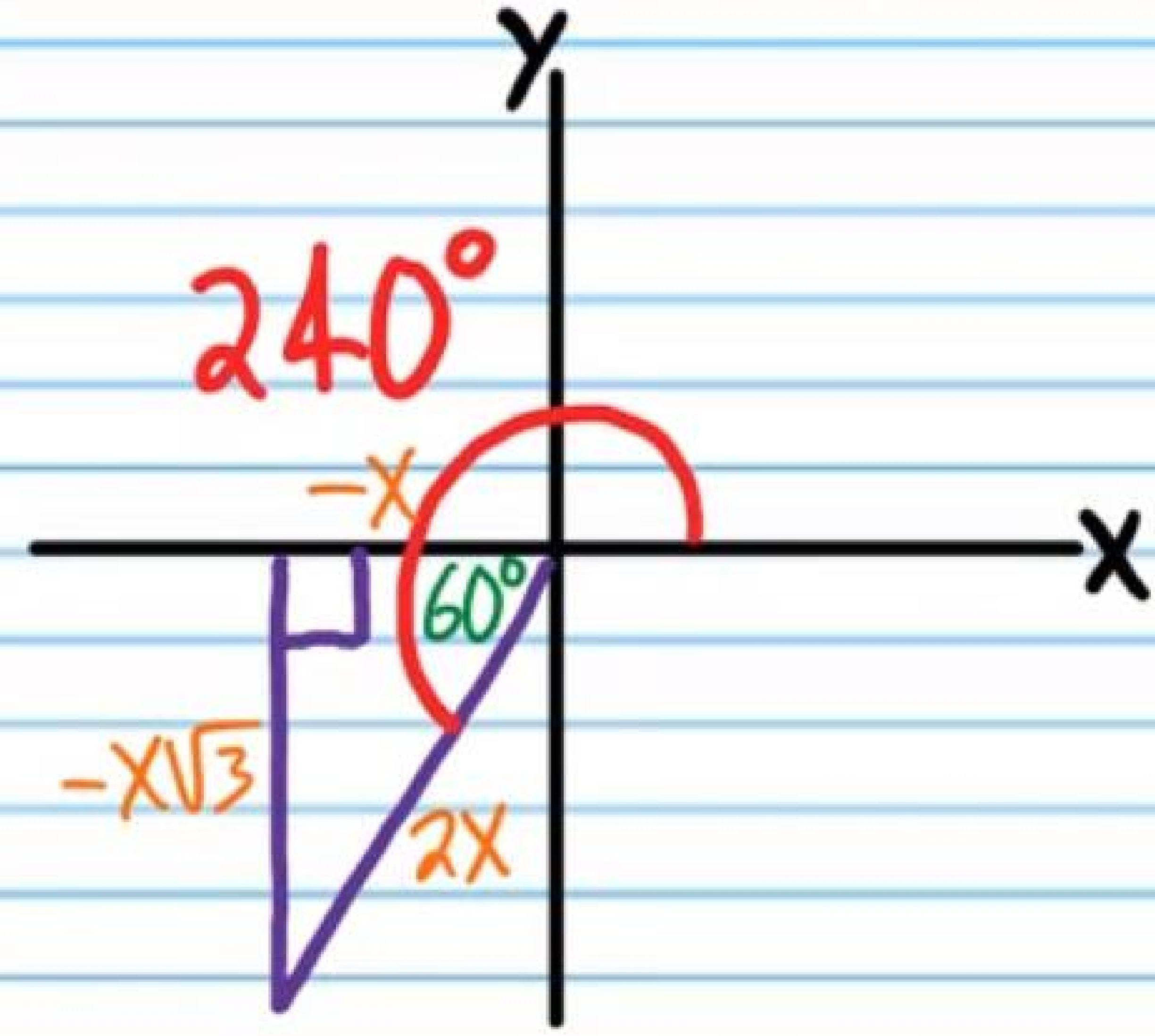
iii) $\tan\theta = \boxed{-\sqrt{3}}$

iv) $\csc\theta = -\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$

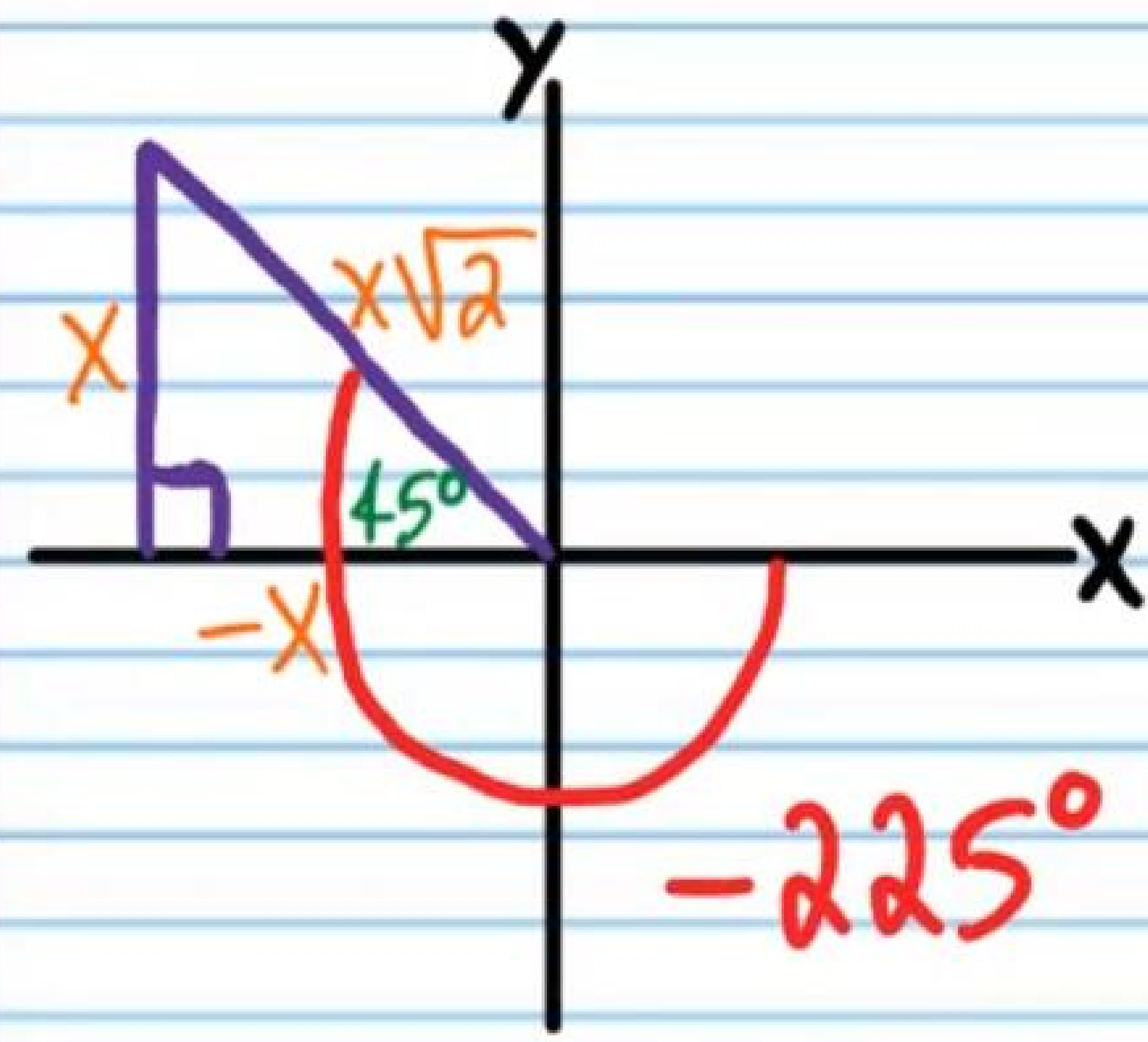
v) $\sec\theta = \boxed{2}$

vi) $\cot\theta = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}}$

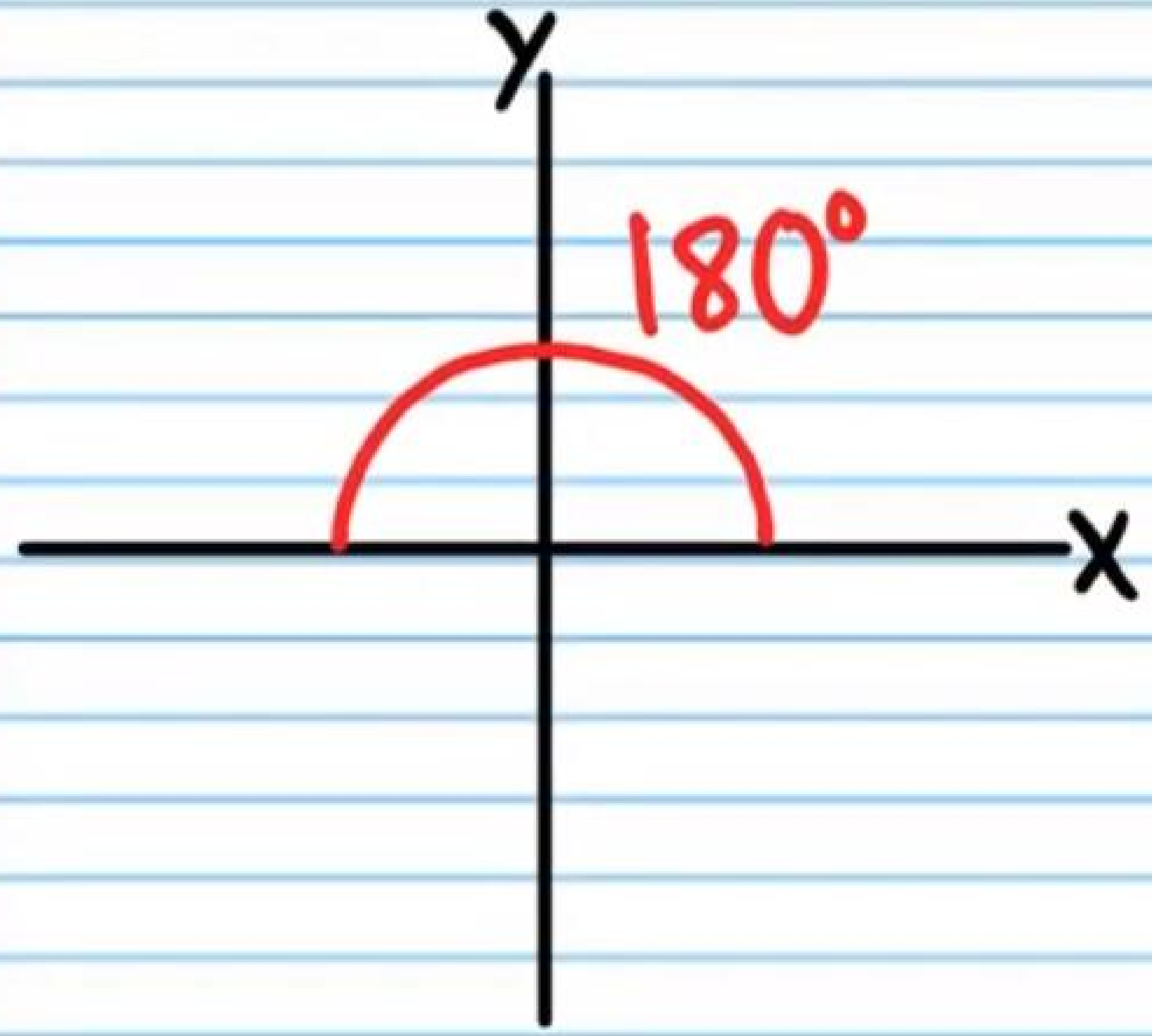
③ $\sec 240^\circ = \boxed{-2}$



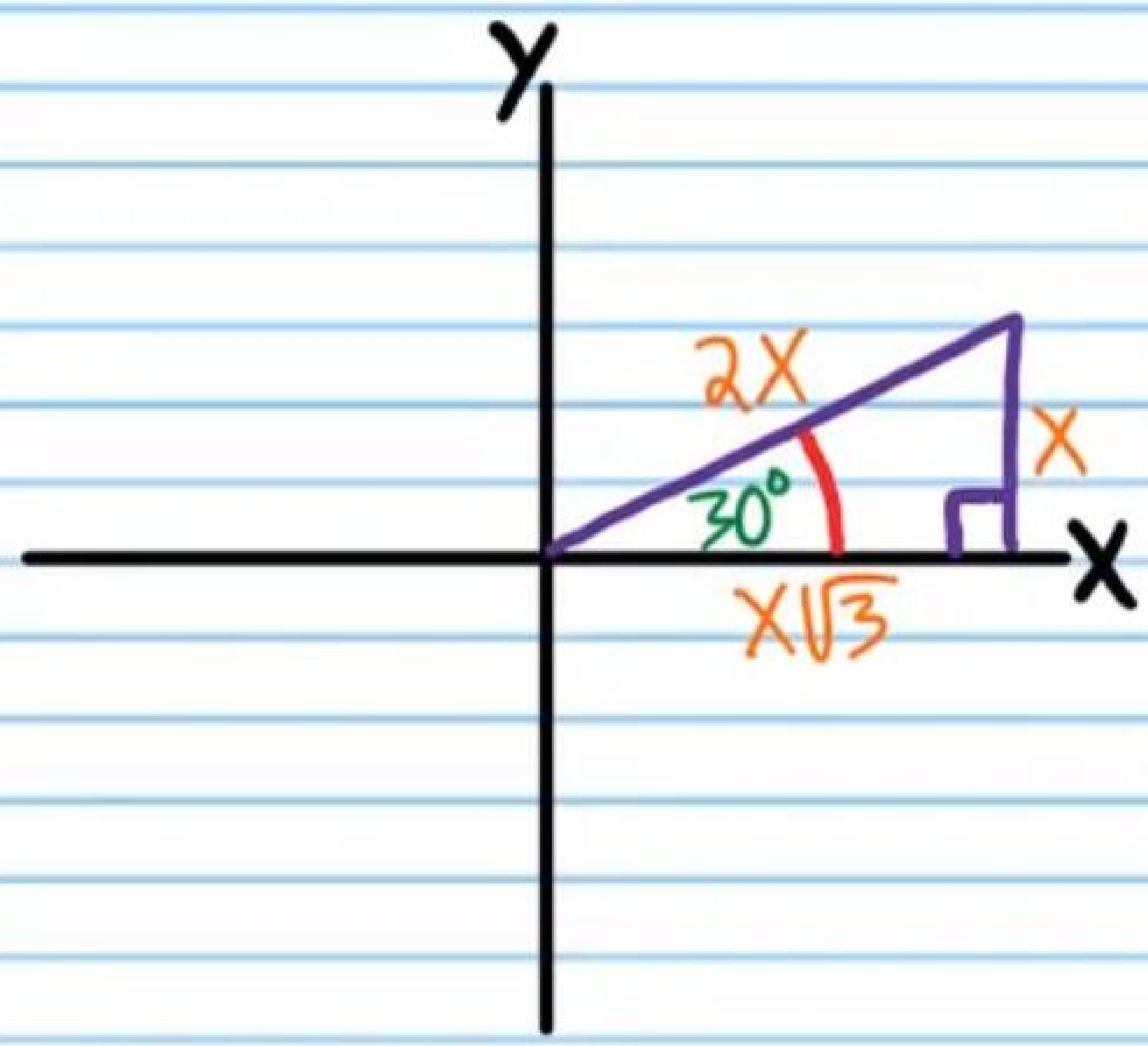
④ $\sin(-225^\circ) = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$



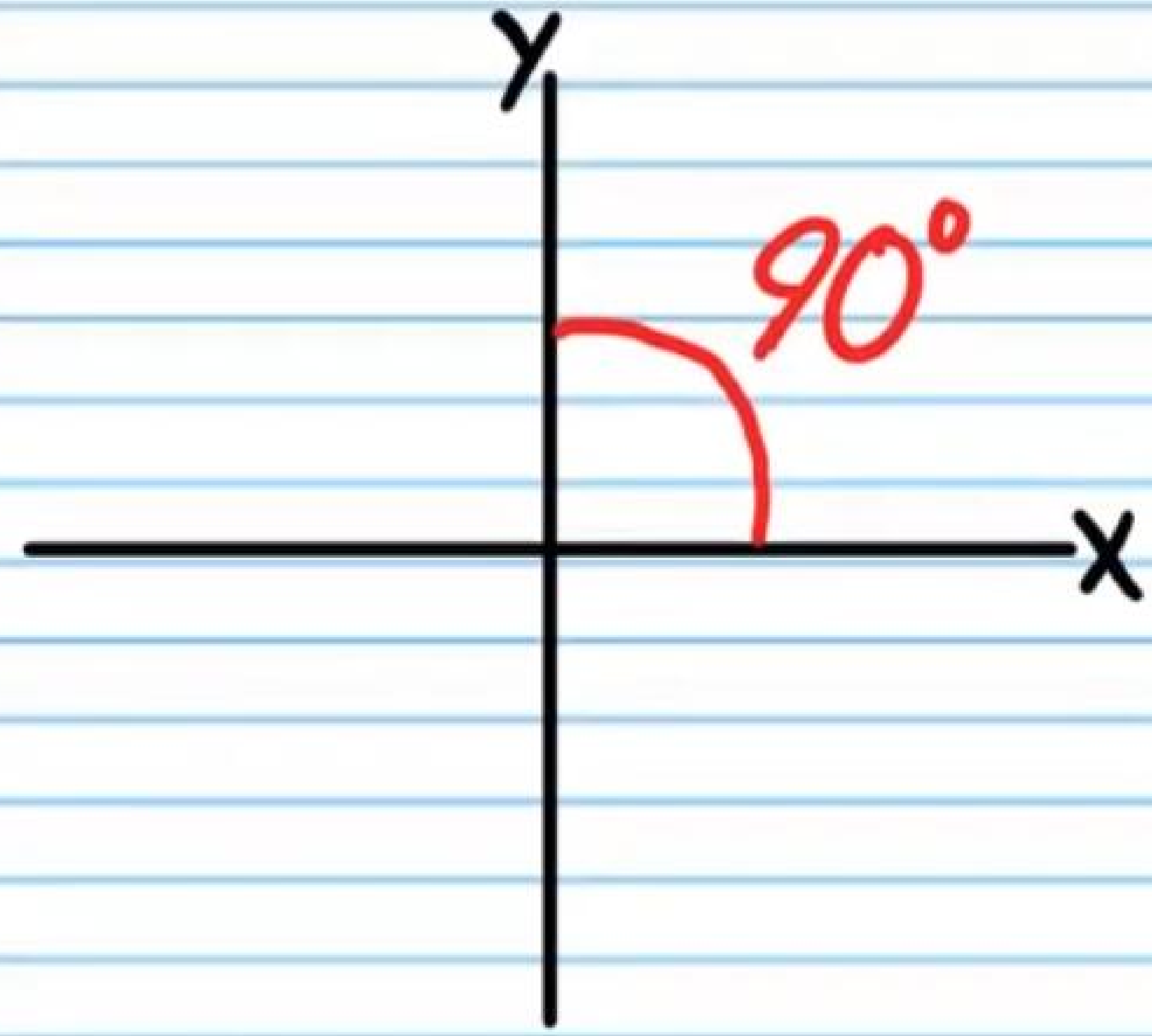
⑤ $\cos 180^\circ = \boxed{-1}$



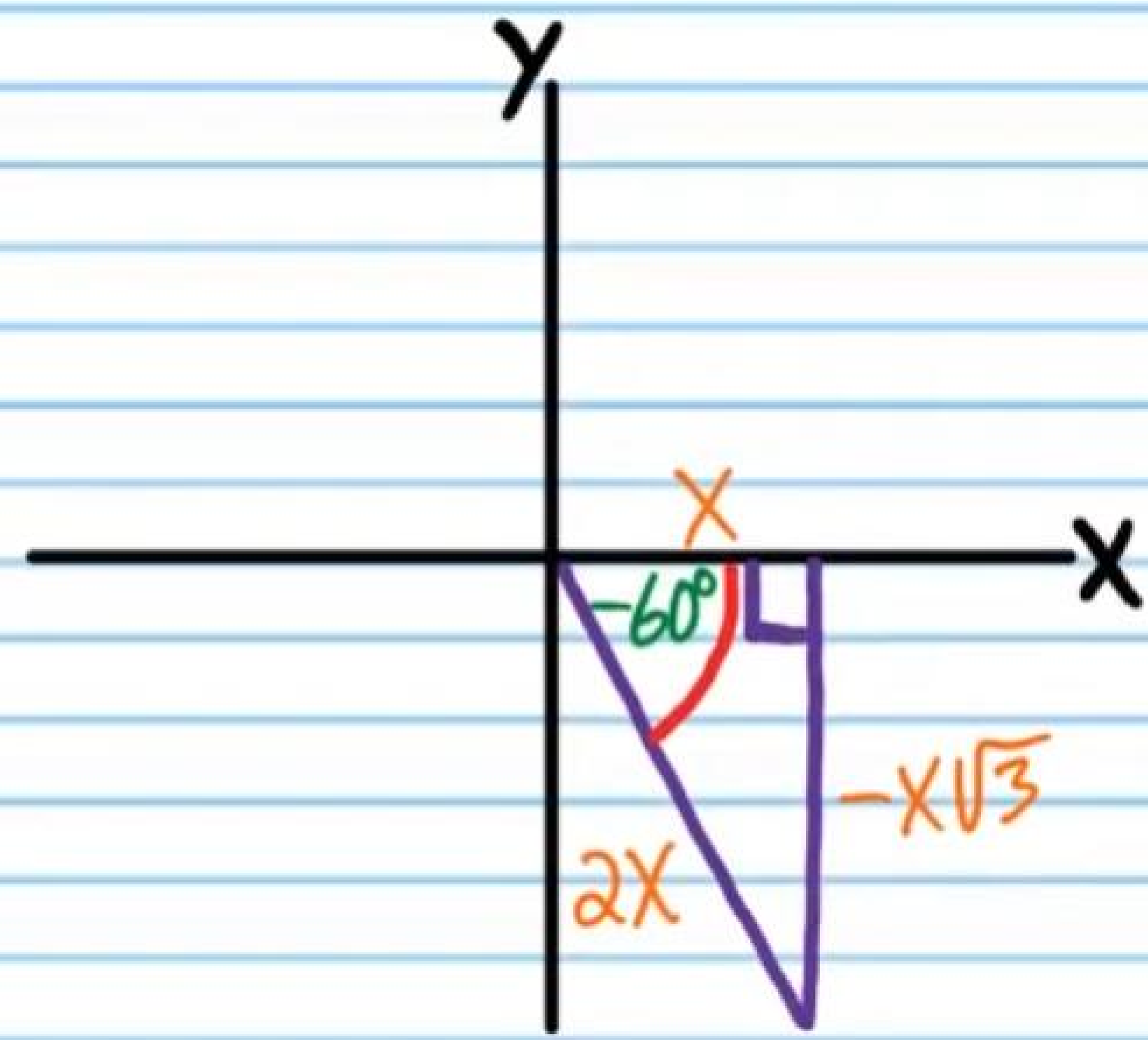
⑥ $\csc 30^\circ = \boxed{2}$



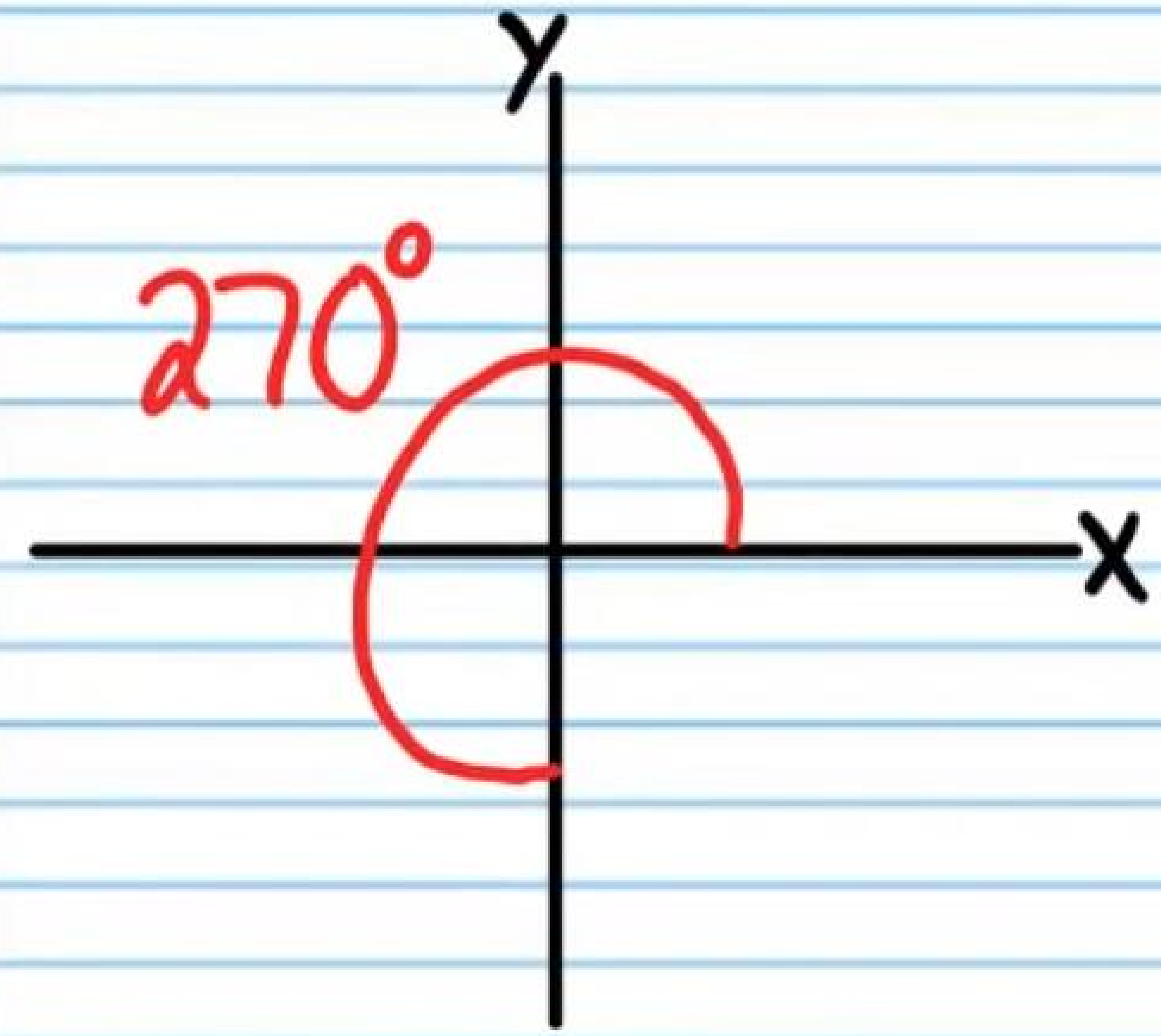
⑦ $\cos 90^\circ = \boxed{0}$



⑧ $\cot(-60^\circ) = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}}$

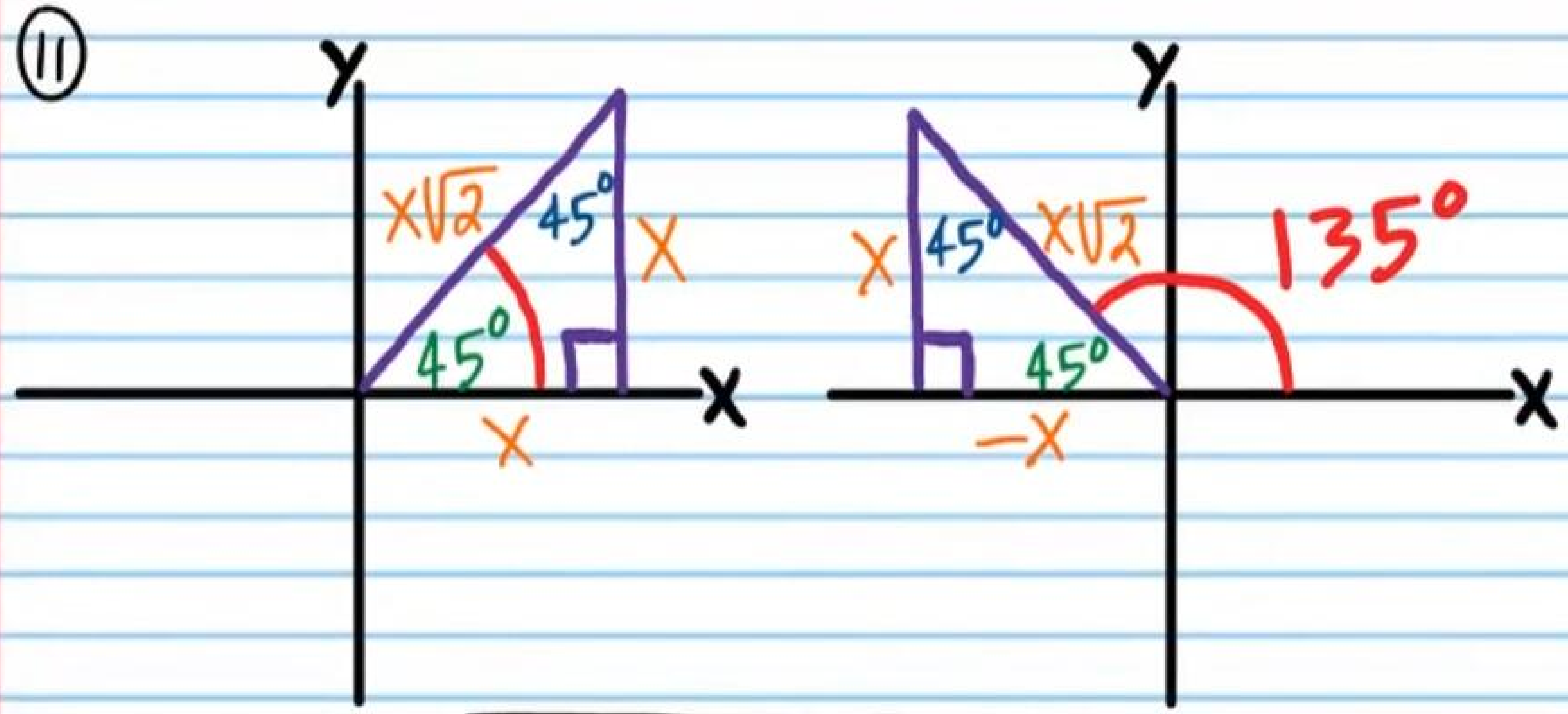


⑨ $\tan 270° = \boxed{\text{Undefined}}$



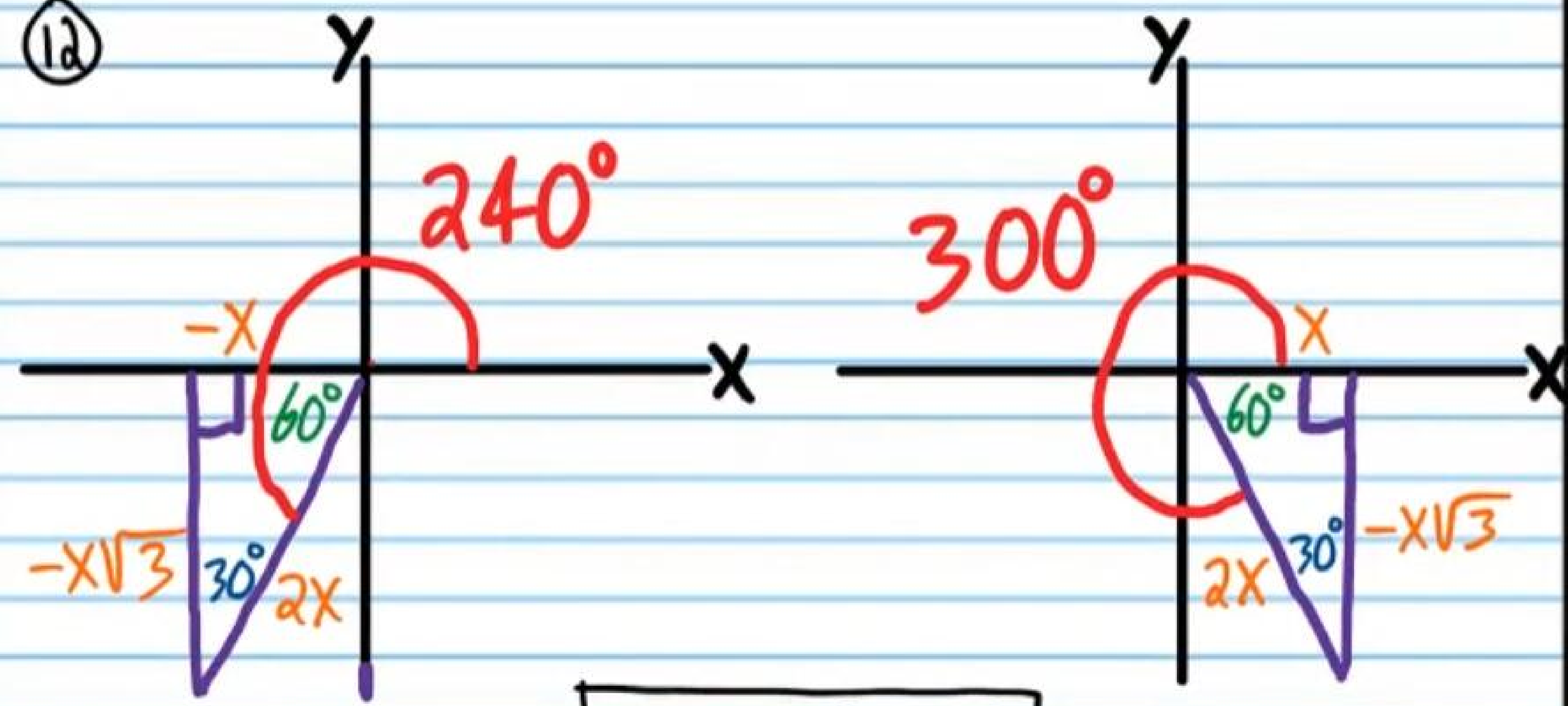
⑩ i) $\boxed{\frac{3}{2}}$ ii) $\boxed{\frac{17}{8}}$

⑪



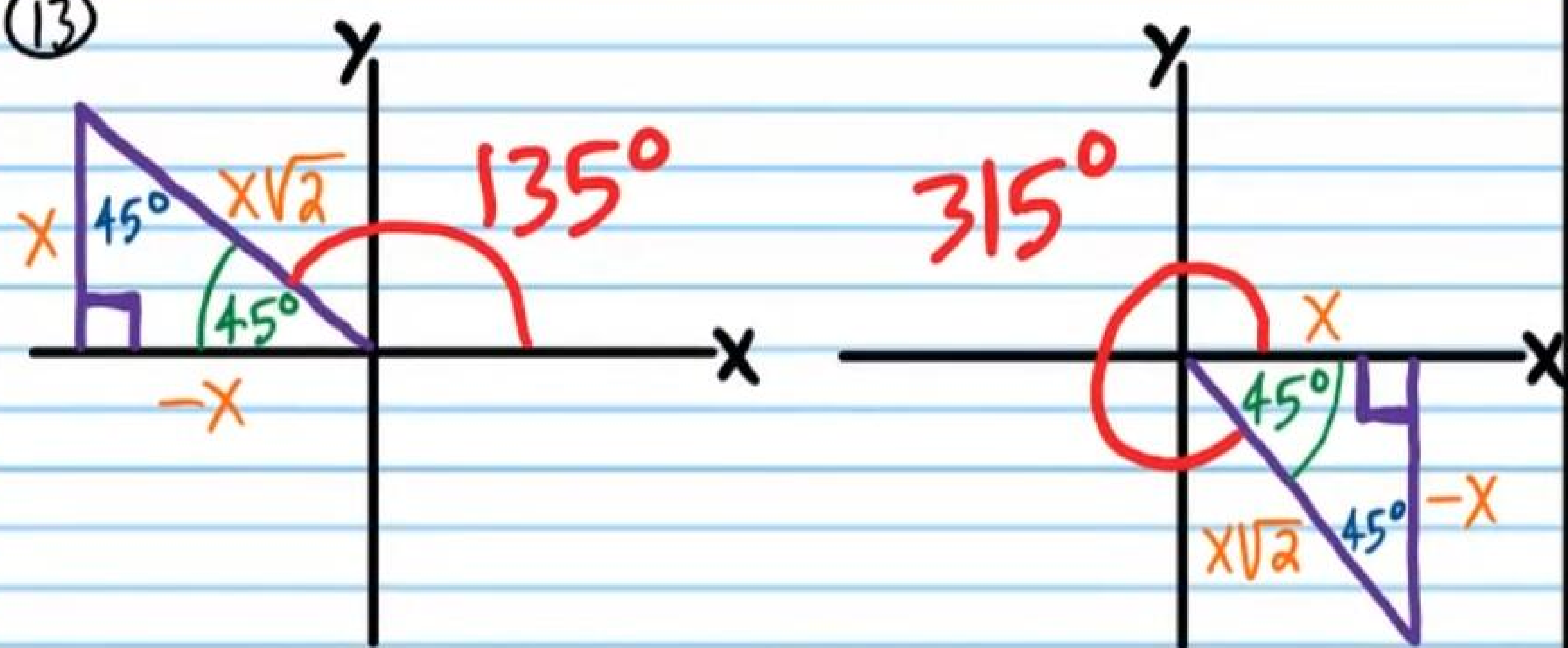
$\boxed{\theta = 45°, 135°}$

⑫



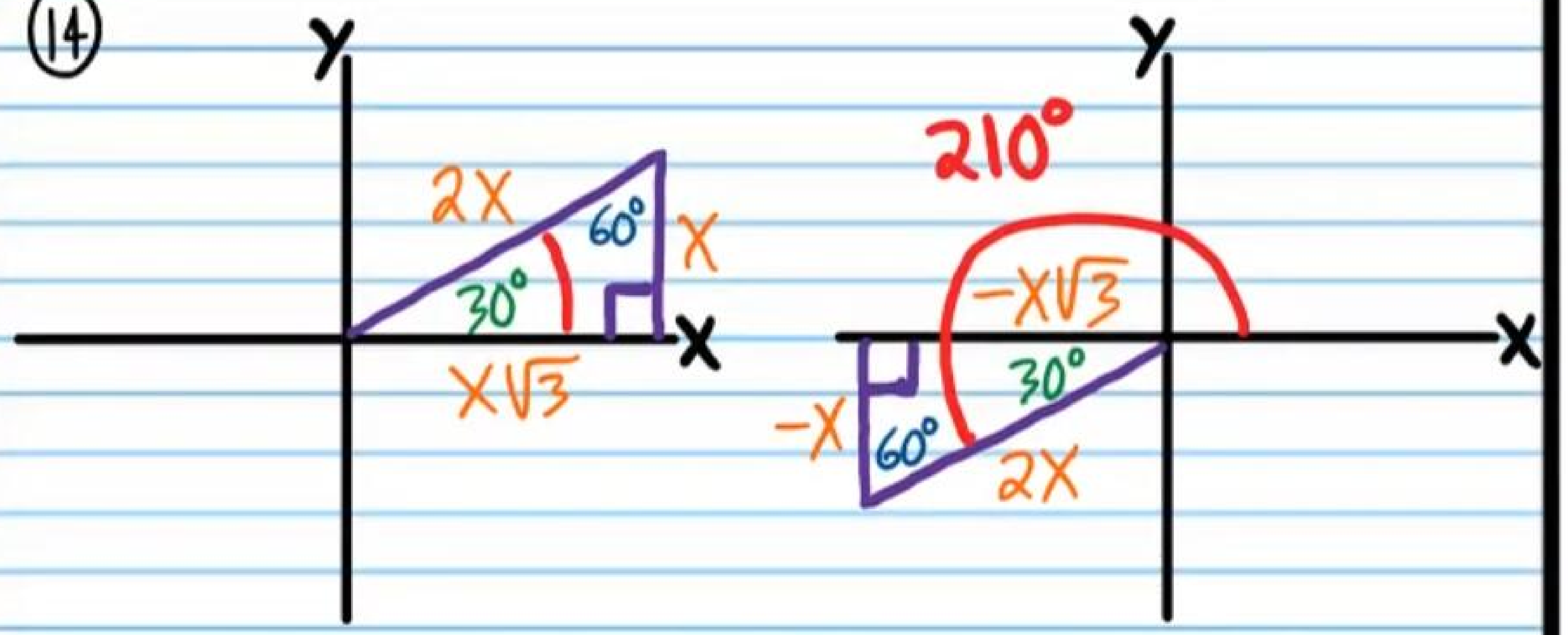
$\boxed{\theta = 240°, 300°}$

⑬



$\boxed{\theta = 135°, 315°}$

(14)

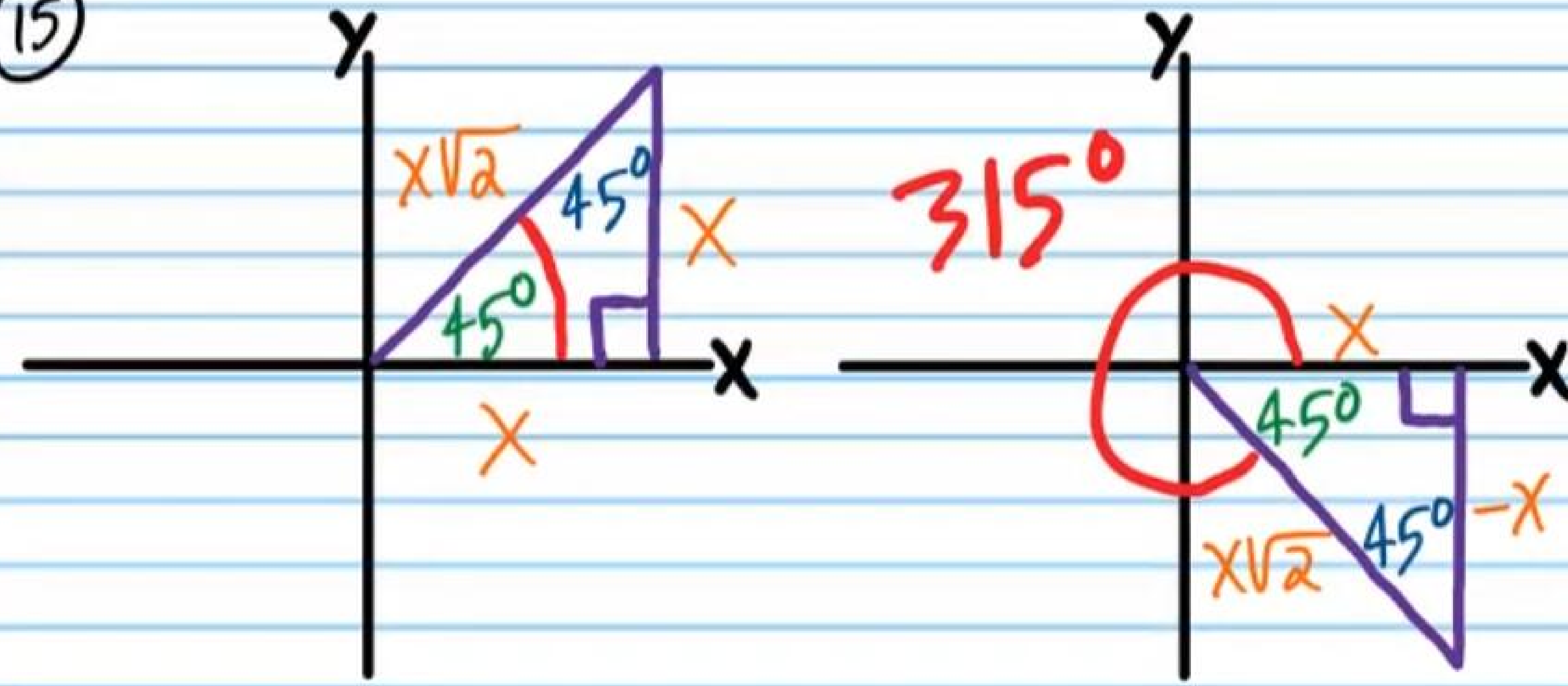


$$\boxed{\begin{array}{l}\theta = 30^\circ + 360n,\ n \text{ is any integer.}\\ \theta = 210^\circ + 360n,\ n \text{ is any integer.}\end{array}}$$

but it is better to combine the two equations into the one below.

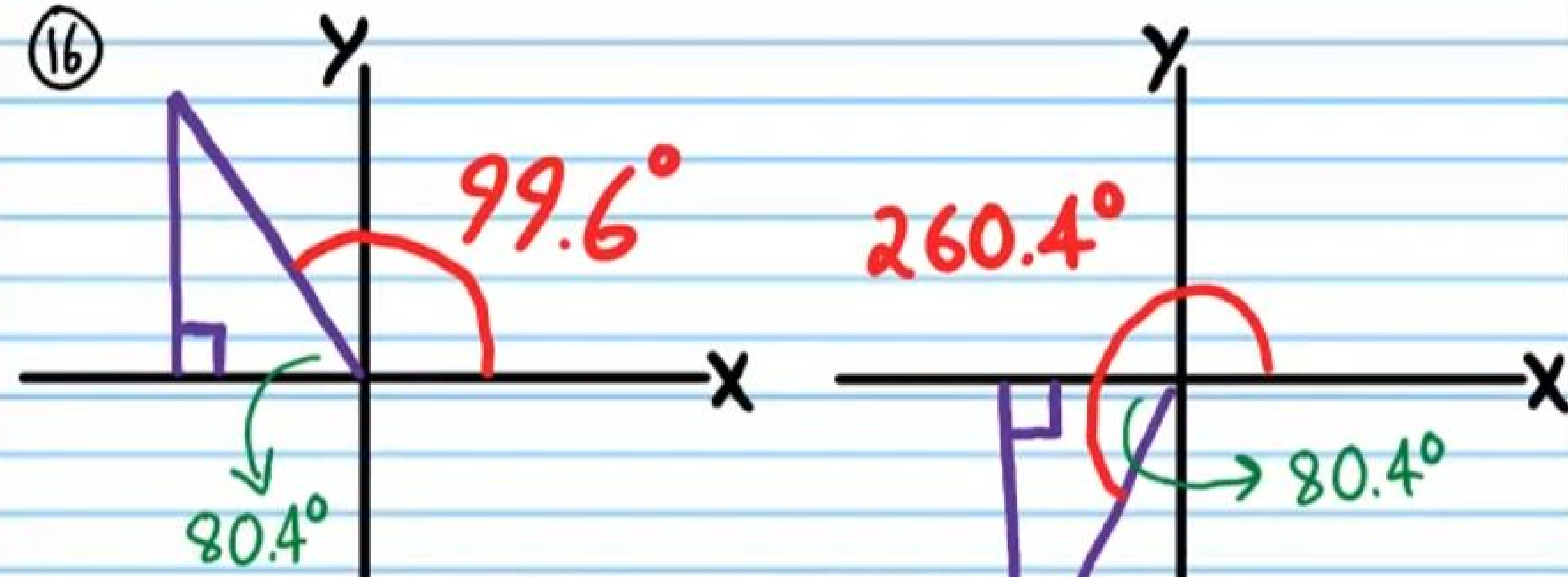
$$\boxed{\theta = 30^\circ + 180n,\ n \text{ is any integer.}}$$

(15)



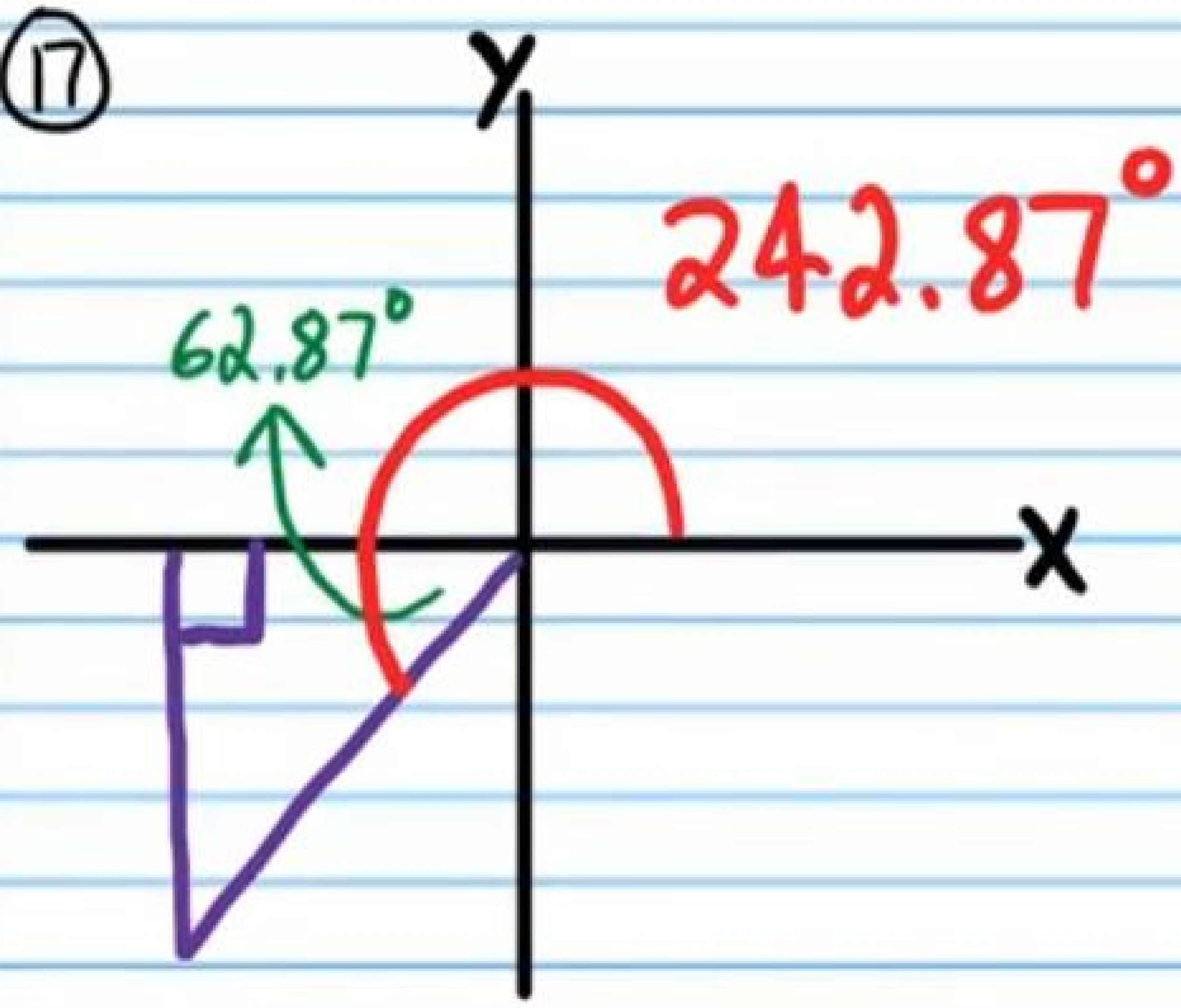
$$\boxed{\begin{array}{l}\theta = 45^\circ + 360n,\ n \text{ is any integer.}\\ \theta = 315^\circ + 360n,\ n \text{ is any integer.}\end{array}}$$

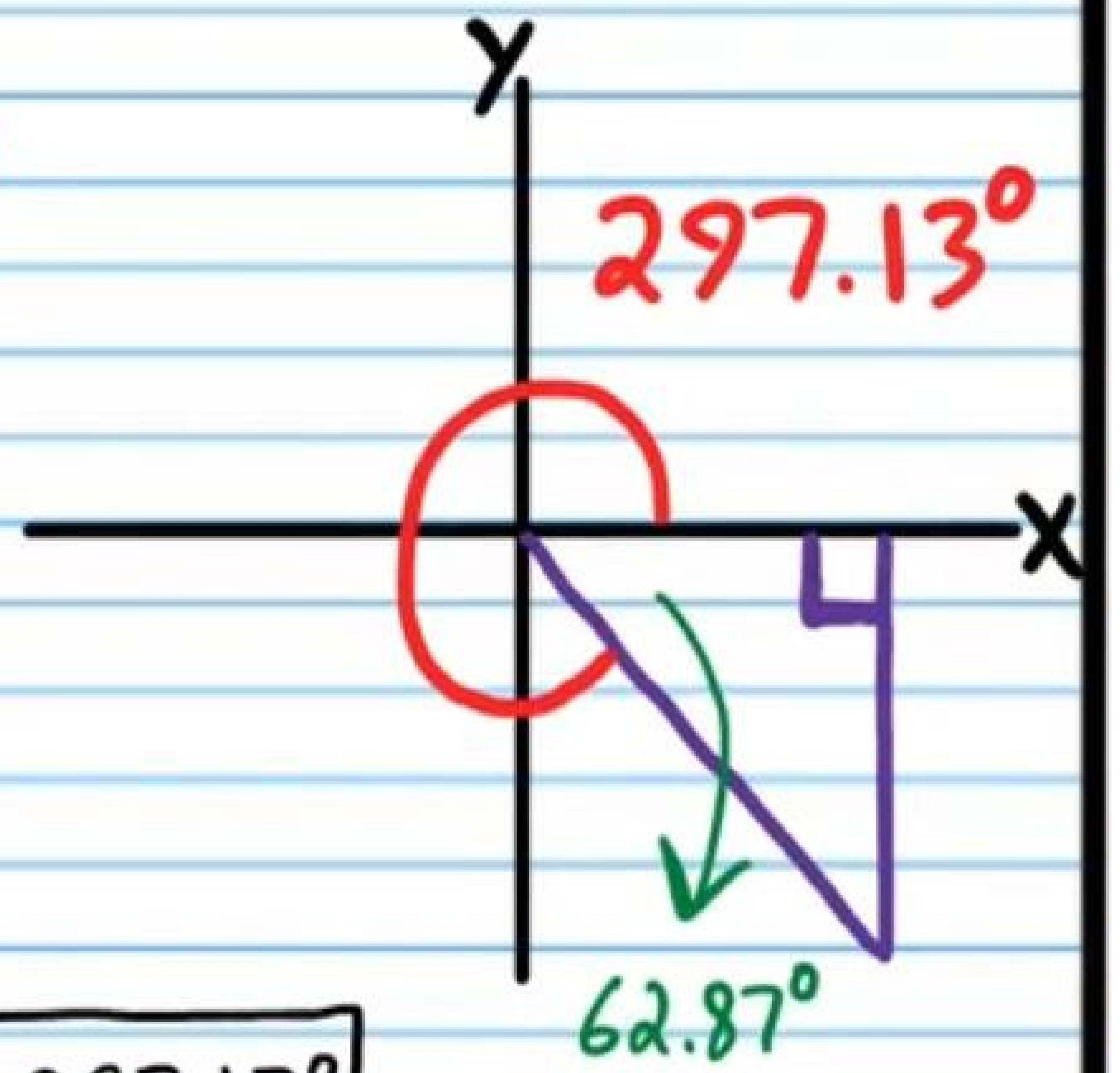
(16)



$$\boxed{\theta \approx 99.6^\circ, 260.4^\circ}$$

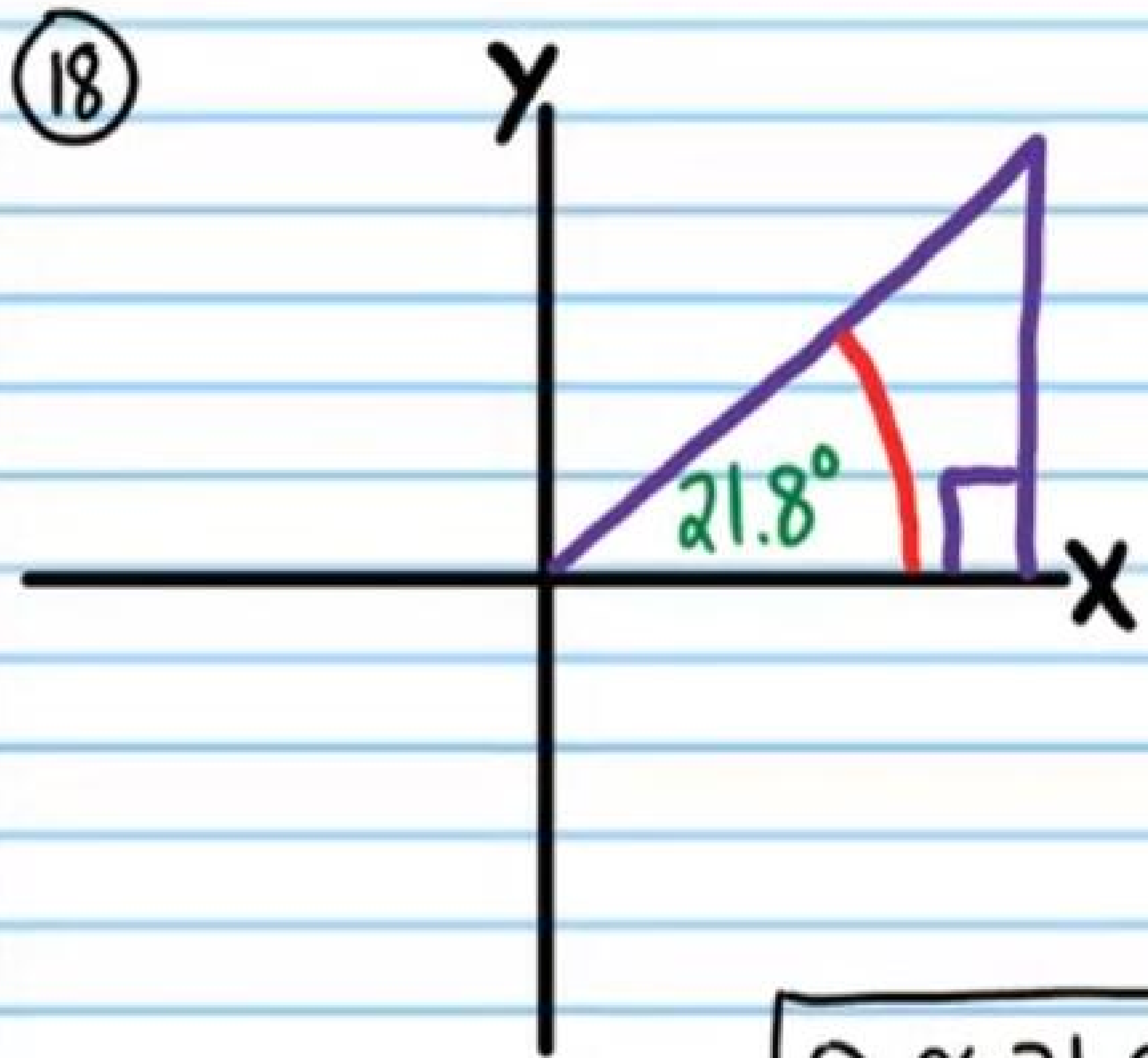
(17)

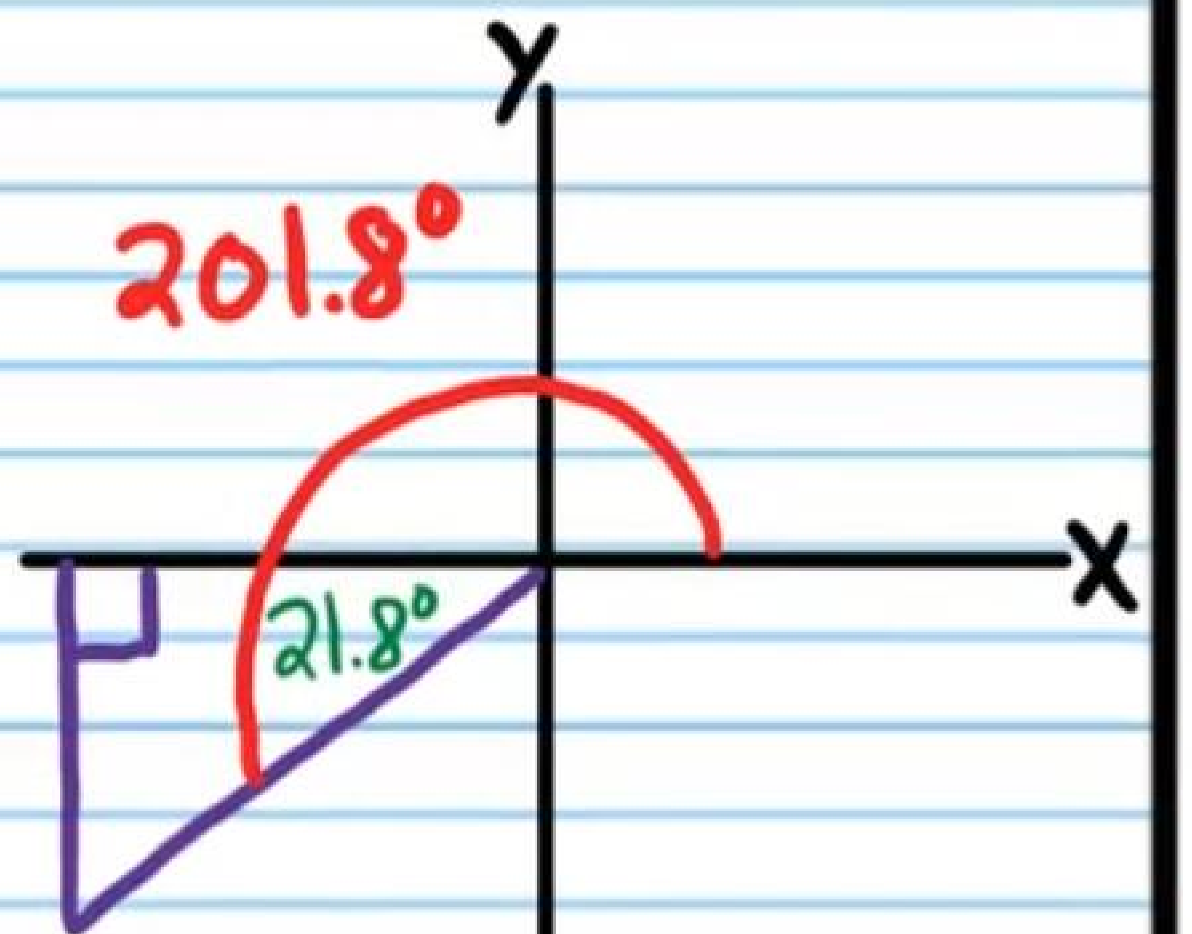




$\boxed{\theta \approx 242.87°, 297.13°}$

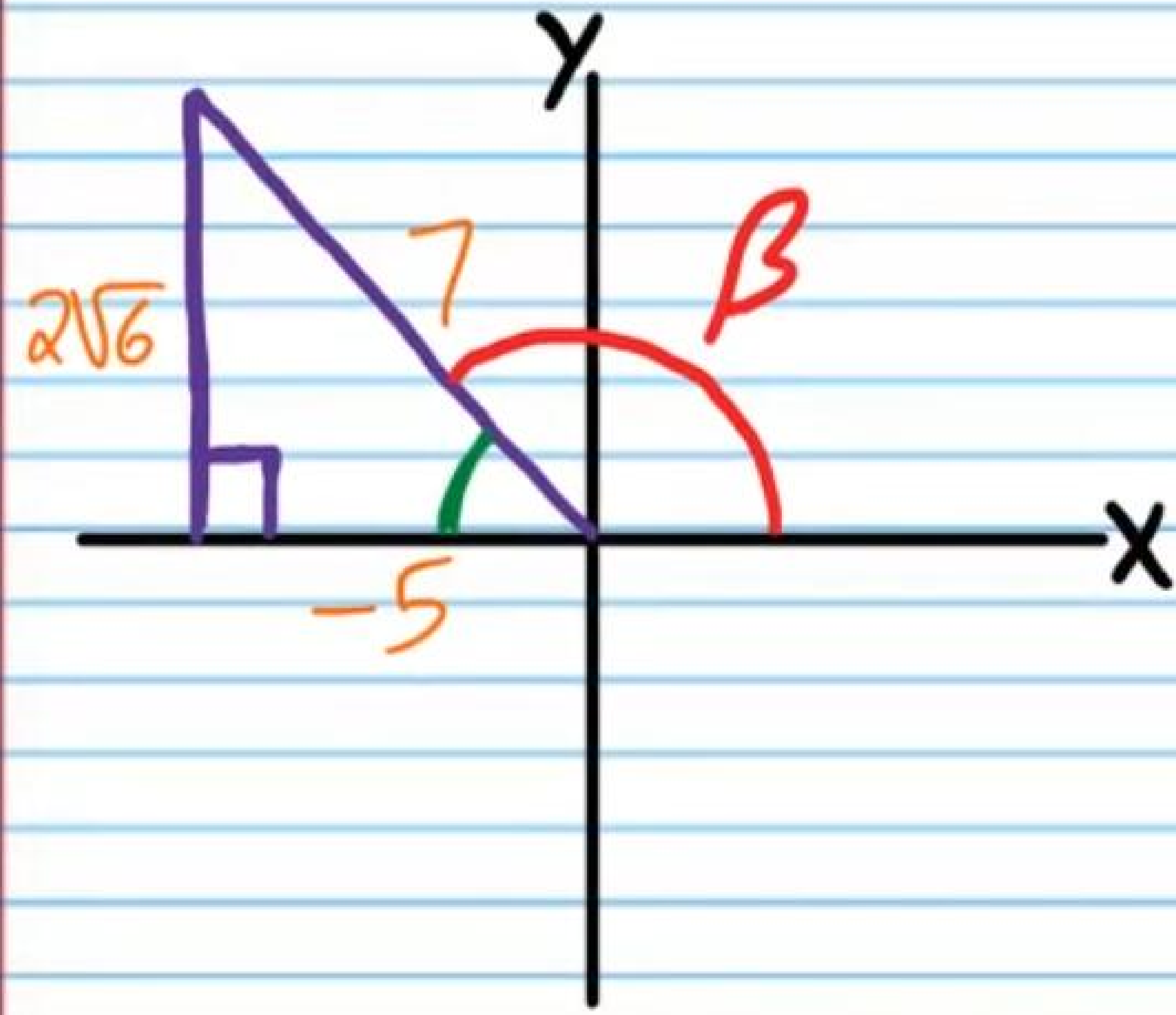
(18)





$\boxed{\theta \approx 21.8°, 201.8°}$

(19) If $\sec\beta = -\frac{7}{5}$, and β terminates in quadrant II, find the values of the five remaining basic trigonometric functions of β.



i) $\cot\beta = -\frac{5}{2\sqrt{6}} = \boxed{-\frac{5\sqrt{6}}{12}}$

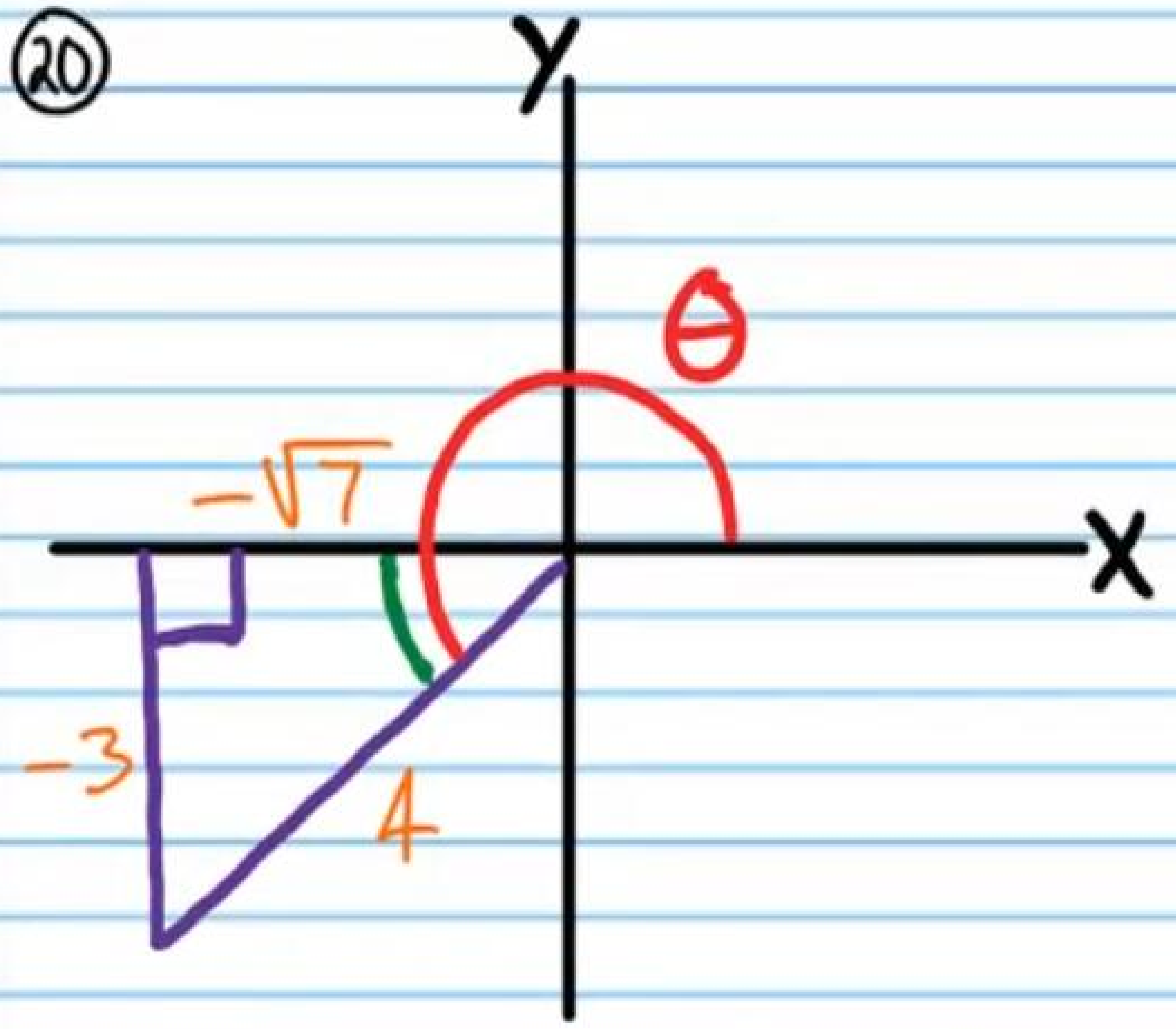
ii) $\cos\beta = \boxed{-\frac{5}{7}}$

iii) $\sin\beta = \boxed{\frac{2\sqrt{6}}{7}}$

iv) $\csc\beta = \frac{7}{2\sqrt{6}} = \boxed{\frac{7\sqrt{6}}{12}}$

v) $\tan\beta = \boxed{-\frac{2\sqrt{6}}{5}}$

(20)



i) $\tan\theta = \frac{3}{\sqrt{7}} = \boxed{\frac{3\sqrt{7}}{7}}$

ii) $\sec\theta = -\frac{4}{\sqrt{7}} = \boxed{-\frac{4\sqrt{7}}{7}}$

iii) $\cot\theta = \boxed{\frac{\sqrt{7}}{3}}$

iv) $\cos\theta = \boxed{-\frac{\sqrt{7}}{4}}$

v) $\sin\theta = \boxed{-\frac{3}{4}}$

(21) Possible Answers: $-661°$, $-301°$, $419°$, $779°$, $1139°$
Only three are necessary.

(22) $\boxed{\theta = 270°}$

(23) $\boxed{\theta = 360n, \text{ n is any integer.}}$

(24) $\boxed{\theta = 90°}$

(25) i) $\sin 329° \approx \boxed{-0.515}$

ii) $\tan 100° \approx \boxed{-5.671}$